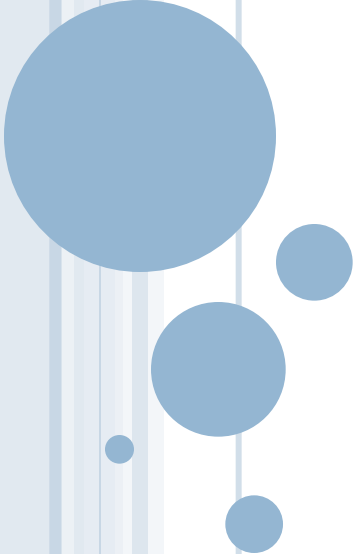




COMPLEX PATHOGENS, HELMINTHS AND IMMUNOLOGY

Paul Thomas

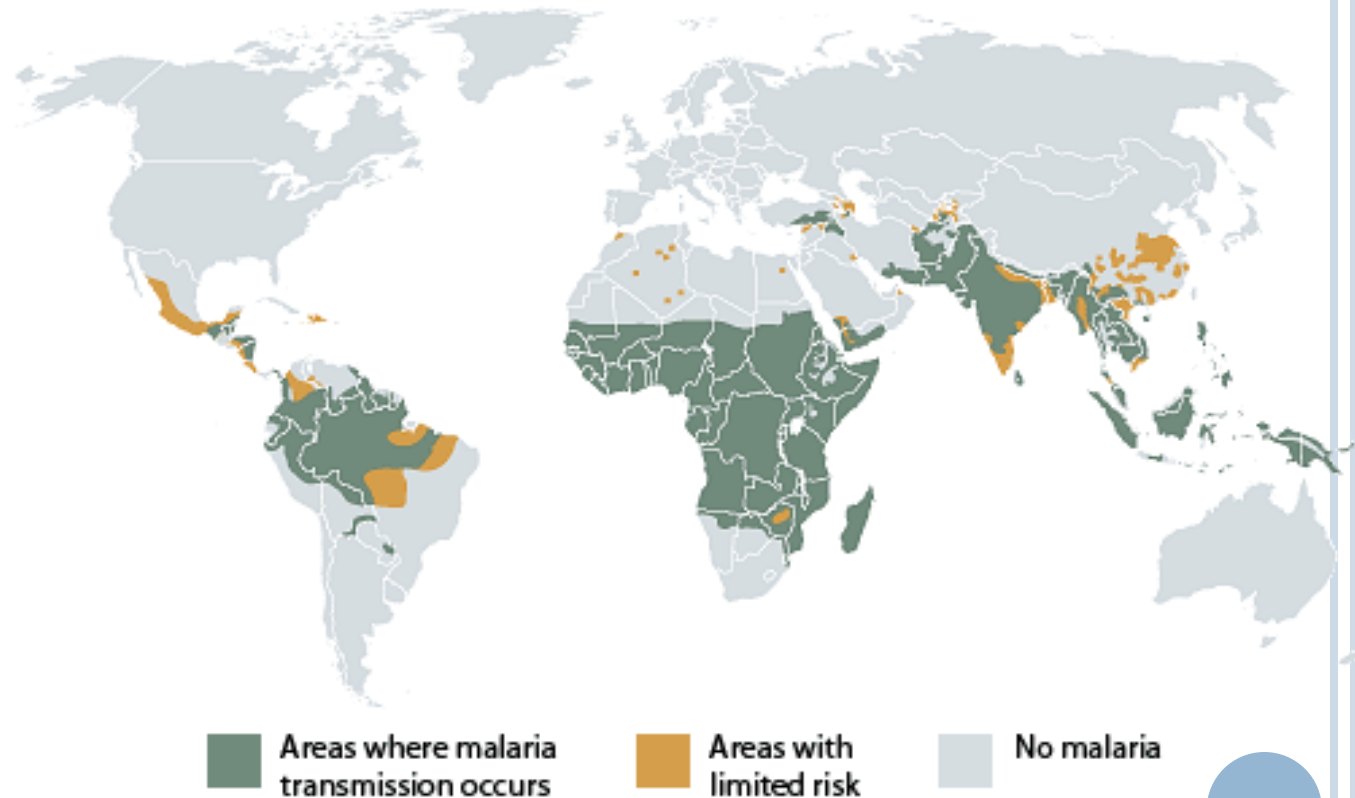
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Fred Hutchinson Cancer Center

MALARIA PREVALENCE

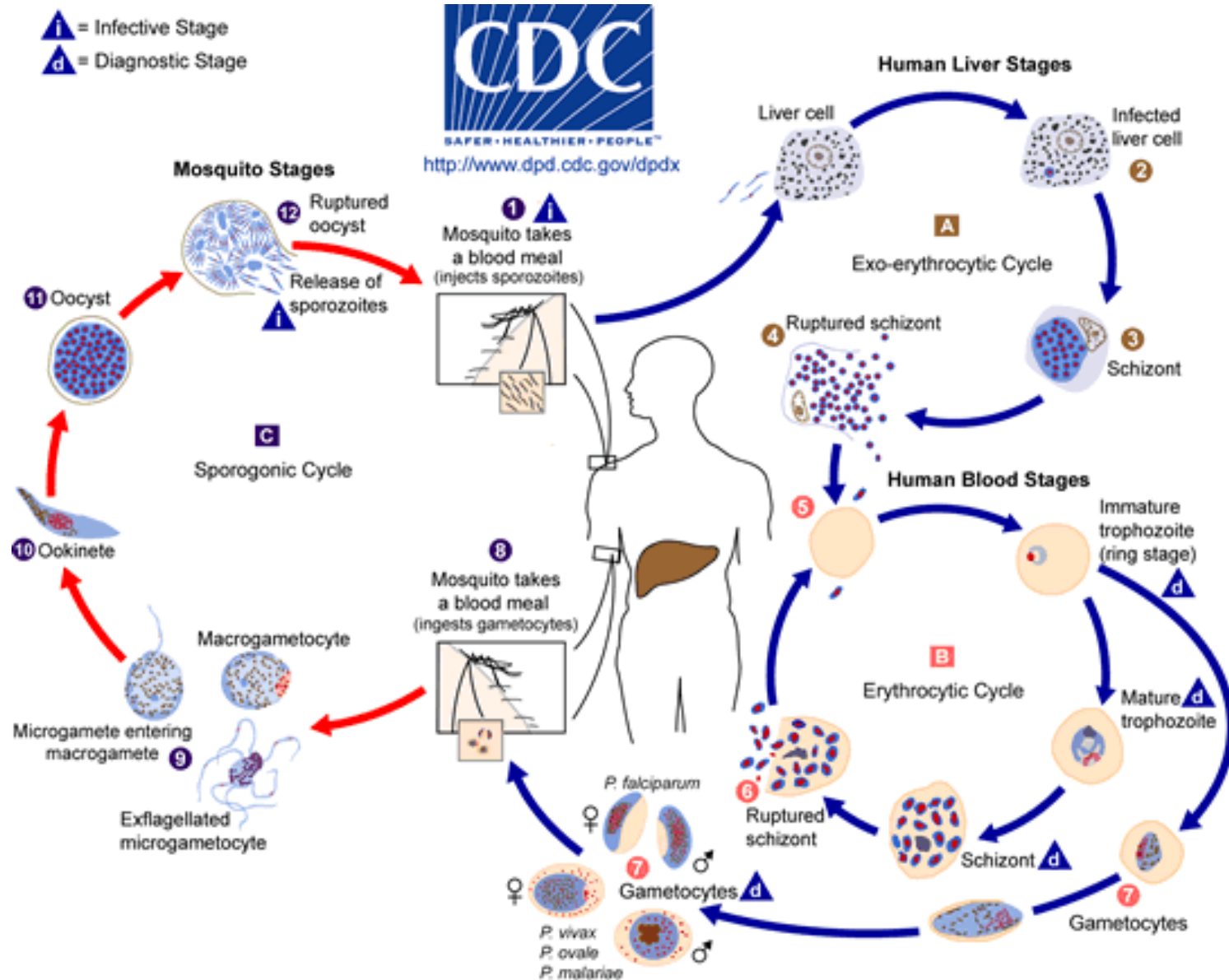
- ~350-500 million infections/year
- 1-2 million deaths/year, most of them among children under 5

Malaria-endemic countries, 2006



Source: International travel and health 2008 page. World Health Organization website.
Available at: www.who.int/ith/en/. Accessed July 31, 2008.

MALARIA LIFE CYCLE



IMMUNE RESPONSES MEDIALTE PLASMODIUM PATHOLOGY

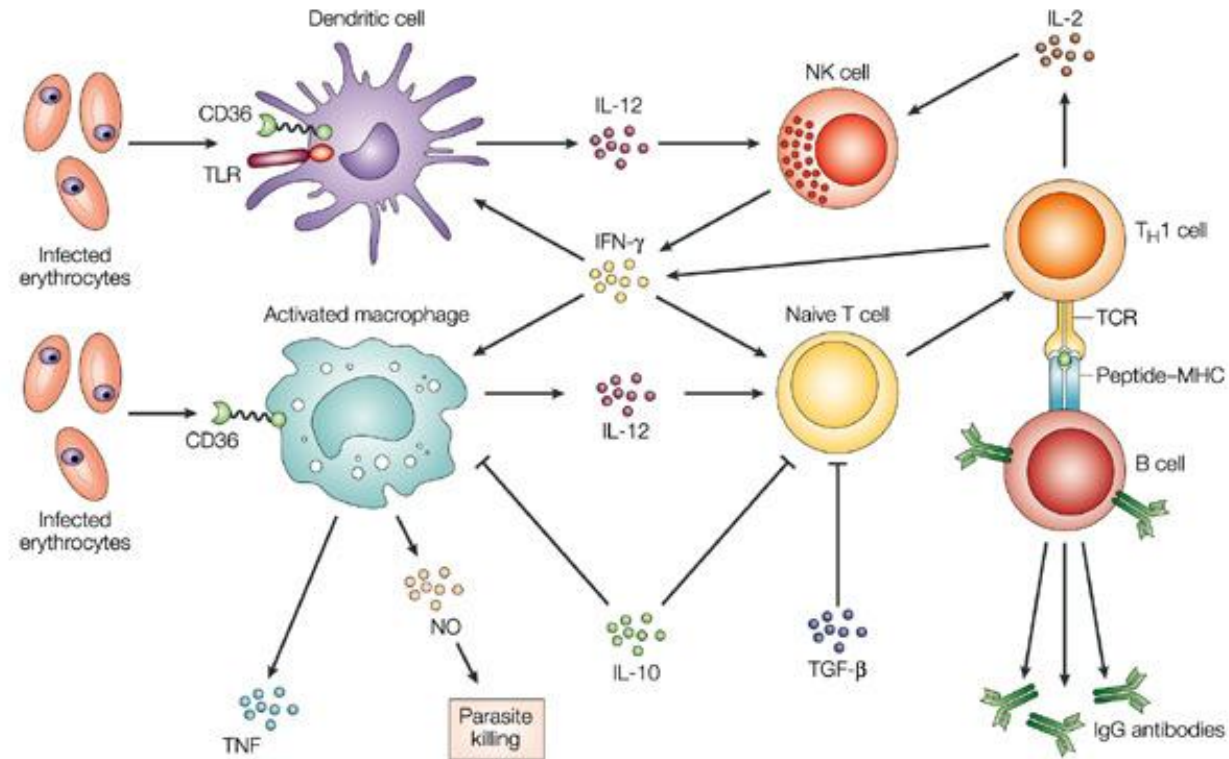
Table 2 | **Malaria products and their bioactivities**

Parasite product	Receptor and cell type	Pathological and cellular effects
<i>Plasmodium falciparum</i> EMP1-family members	ICAM1, VCAM1, CD36, thrombospondin, E-selectin, chondroitin sulphate A, hyaluronic acid and CD31 on endothelial cells and trophoblast cells; CD36 on DCs	Binding directs parasite to the brain, placenta and possibly other target organs; CD36 engagement proposed to suppress DC and macrophage activation
GPI	TLR2, TLR4 and/or possibly C-type lectins on several cell types, including DCs, macrophages, endothelial cells and adipocytes; CD1d and V α 14-V β 8 TCR on NKT cells	Induces widespread expression of genes encoding pro-inflammatory proteins (including TNF, IL-1, IL-6, IL-12, iNOS, ICAM1, VCAM1); activates NKT cells; induces T _H 1- or T _H 2-cytokine production
Haemozoin	TLR9 on DCs	Contradictory reports: both T _H 1- and T _H 2-cell activities; induces and inhibits DCs; suppresses macrophages; induces IL-10 production; broadly immunosuppressive
Unknown ligands	NKC-encoded receptors on NK and NKT cells	Activates NK cells; induces IFN- γ production; regulates balance of T _H 1 and T _H 2 cytokines produced by NKT cells
Isopentenyl pyrophosphate	$\gamma\delta$ TCRs	Activates $\gamma\delta$ T cells; induces IFN- γ production
Protein antigens	Diverse TCRs on CD4 ⁺ and CD8 ⁺ T cells	Activates $\alpha\beta$ T cells; induces T _H 1- or T _H 2-cytokine production
Unknown sugar(s)	MBL in plasma	Possible binding provides protection; low levels of MBL are associated with disease

DC, dendritic cell; EMP1, erythrocyte membrane protein 1; E-selectin, endothelial-cell selectin; GPI, glycosylphosphatidylinositol; ICAM1, intercellular adhesion molecule 1; IFN- γ , interferon- γ ; IL, interleukin; iNOS, inducible nitric-oxide synthase; MBL, mannose-binding lectin; NK, natural killer; NKC, natural killer complex; NKT, natural killer T; TCR, T-cell receptor; T_H, T helper; TLR, Toll-like receptor; TNF, tumour-necrosis factor; VCAM1, vascular cell-adhesion molecule 1.

TYPICAL “Th1-TYPE” IMMUNITY CONTRIBUTES TO MALARIA PROTECTION AND PATHOLOGY

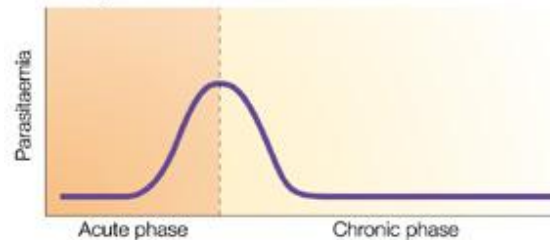
- In infected mice, antigen is presented in the spleen where Th1 cells regulate innate and adaptive immune responses, including stimulating anti-parasite antibody and effector mechanisms such as ROI and RNI



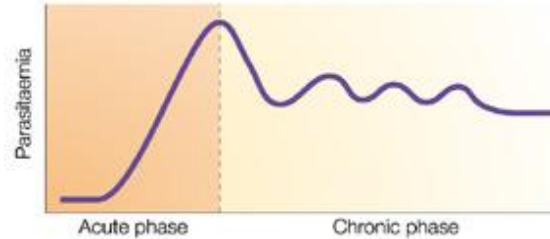
IMMUNE MECHANISMS TO CONTROL MALARIA

- Antibodies block invasion of sporozoites into liver cells
- Interferon- (IFN-) and CD8⁺ T cells inhibit parasite development in hepatocytes
- Antibodies block invasion of merozoites into erythrocytes
- Antibodies prevent sequestration of infected erythrocytes by preventing binding to adhesion molecules on the vascular endothelium
- IFN- and CD4⁺ T cells activate macrophages to phagocytose intra-erythrocytic parasites and free merozoites
- Antibodies neutralize parasite glycosylphosphatidylinositol and inhibit induction of the inflammatory cytokine cascade
- Antibodies mediate complement-dependent lysis of extracellular gametes, and prevent fertilization of gametes and the development of zygotes

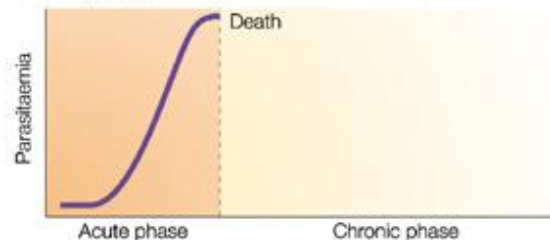
a Wild-type mice



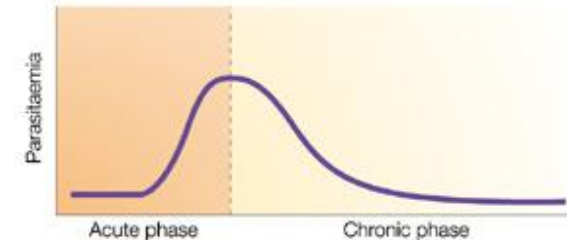
b CD4⁺ T-cell-depleted mice



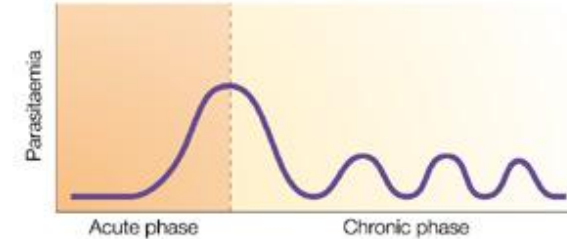
c IFN-γ-deficient mice



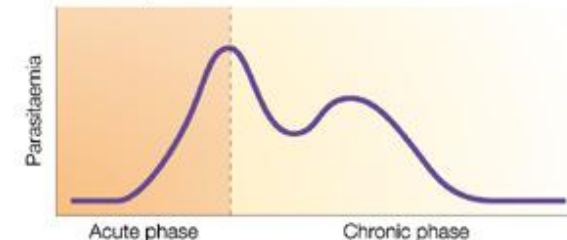
d γδ T-cell-deficient mice



e B-cell-depleted or -deficient mice



f NK-cell-depleted mice



FATAL DISEASE IN MALARIA INFECTION

Table 1 | **Severe and fatal disease syndromes in malaria**

Syndrome	Clinical features	Possible sequence or mechanism of disease
Cerebral malaria	Sustained impaired consciousness, coma, long-term neurological sequelae	Cerebral parasite sequestration; bioactive GPI; pro-inflammatory cytokine cascade; endothelial-cell activation; natural killer T-cell activation; T_H1/T_H2 -cell balance; chemokine production; monocyte, macrophage and neutrophil recruitment; platelet and fibrinogen deposition; $CD4^+$, $CD8^+$ and $\gamma\delta$ T-cell involvement; IFN- γ production; neurological metabolic derangements; possibly hypoxia
Placental malaria	Placental insufficiency, low birth weight, premature delivery, loss of fetus	<i>Plasmodium falciparum</i> EMP1-mediated binding to placental endothelium and syncytiotrophoblast through chondroitin sulphate A and hyaluronic acid; cytokine production; chemokine-mediated recruitment and infiltration of monocytes; intravascular macrophage differentiation
Severe malarial anaemia	Pallor, lethargy, haemoglobin level of 4–6 g per 10 ml	Erythropoietic suppression by toxins and cytokines; increased RBC destruction, owing to parasitization, RBC alterations, complement and immune complex or antigen deposition, erythrophagocytosis, splenic hyperphagism, $CD4^+$ T cells, T_H1/T_H2 cytokine balance (TNF and IFN- γ versus IL-10)
Metabolic acidosis	Respiratory distress, deep breathing (Kussmaul breathing), hypovolaemia	Molecular mechanisms unknown. Possibly widespread parasite sequestration; bioactive toxins; increased vascular permeability; reduced tissue perfusion; anaemia; pulmonary airway obstruction; hypoxia; increased host glycolysis; repressed gluconeogenesis. Some overlap with shock-like syndrome
Shock-like syndrome (systemic inflammatory-response-like syndrome)	Shock, haemodynamic changes, impaired organ perfusion, disseminated intravascular coagulation	Bioactive toxins; T_H1 cytokines; acute-phase reactants

EMP1, erythrocyte membrane protein 1; GPI, glycosylphosphatidylinositol; IFN- γ , interferon- γ ; IL-10, interleukin-10; RBC, red blood cell; T_H , T helper; TNF, tumour-necrosis factor.

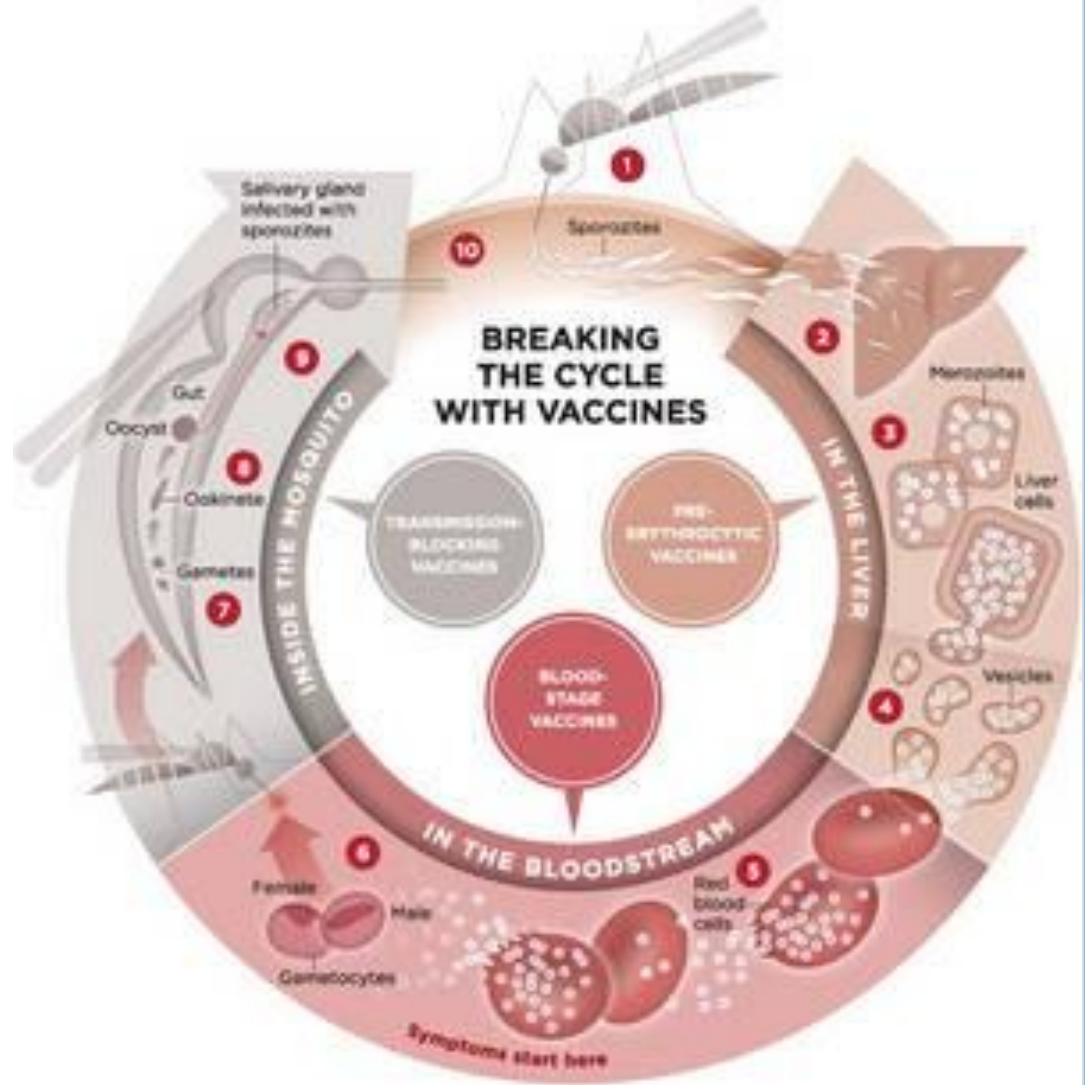
VECTOR CONTROL

- Vector control efforts range from basic bed nets, to spraying insecticides externally and on house walls, to more sophisticated “vector engineering” efforts to produce malaria-resistant mosquitoes, among many others
- Math modeling of infectious spread has led to some hypotheses about which of these methods are the most effective (bed nets, house wall spraying) and which are unlikely to be effective (releasing resistant mosquitoes)



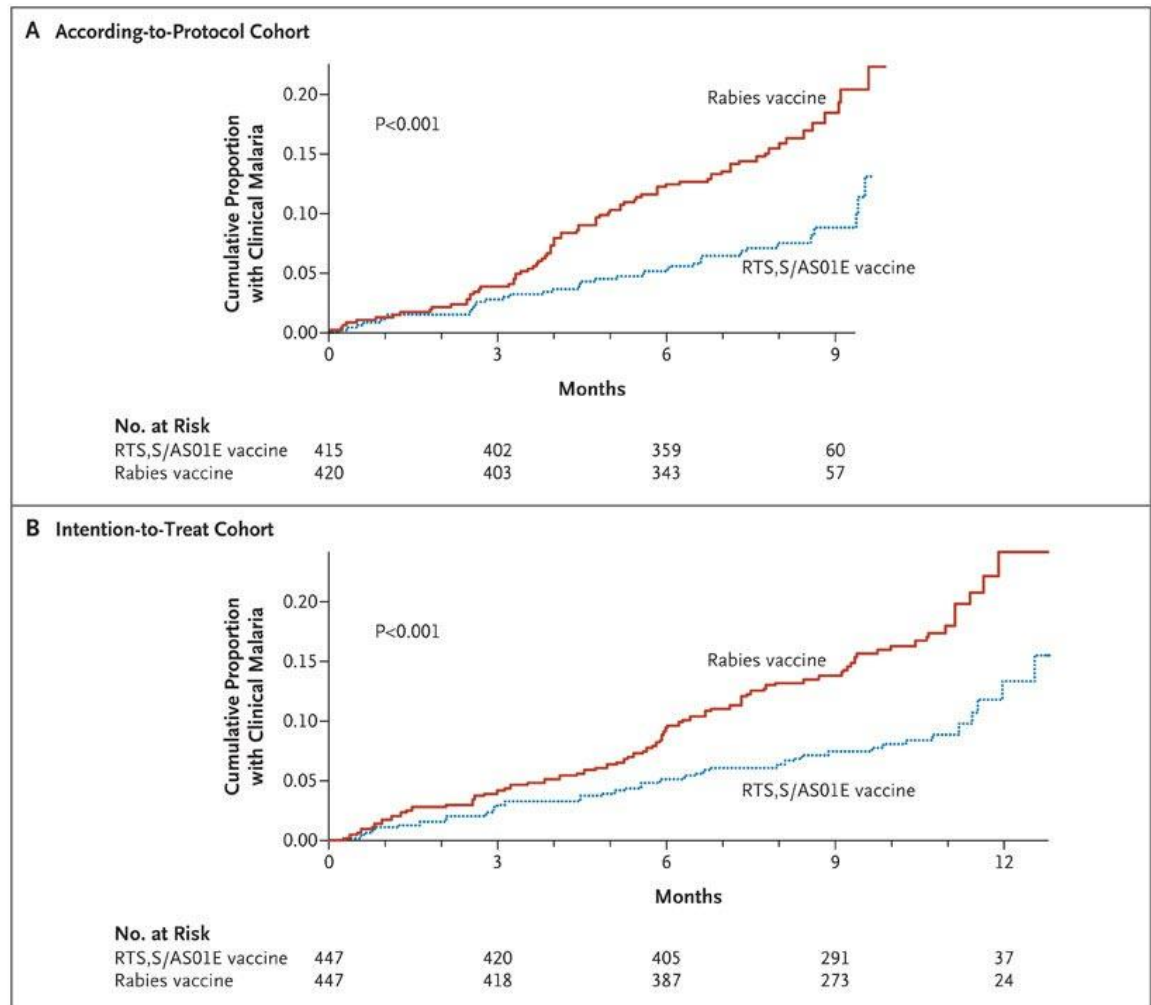
MALARIA VACCINE TARGETS

- Three types of vaccines have been proposed, with variations in each group:
 - Pre-erythrocytic vaccines: the only truly “sterilizing” protection, but hard to generate enough antibody immunity (to prevent any infection) or CD8 immunity (to clear every single infected liver cell)
 - Blood-stage vaccine: designed to enhance clearance of infected red blood cells, therapeutic but not sterilizing
 - Gametocyte vaccines: Potentially strong antigen candidates and immune complexes can be carried to the mosquito—“altruistic vaccine”



RTS,S VACCINE APPROVED FOR STAGE III IN INFANTS

- The vaccine candidate farthest along is RTS,S, pre-erythrocytic vaccine against the circumsporozoite protein
- Mechanism is presumed to be antibody, but cellular responses have been shown
- Vaccine is adjuvanted and protein is linked to hepatitis B antigen



RTS,S SHOWED MODEST EFFICACY IN INFANTS

- ~30% efficacy shown in latest trial
- Generally viewed as disappointing, but still moving forward (previous trial had ~61% efficacy, but was much smaller and in a different transmission area)

N ENGL J MED 367;24 NEJM.ORG DECEMBER 13, 2012

The NEW ENGLAND JOURNAL of MEDICINE

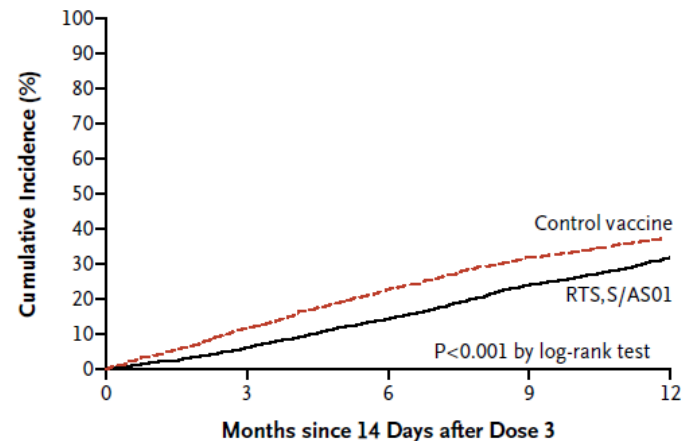
ORIGINAL ARTICLE

A Phase 3 Trial of RTS,S/AS01 Malaria Vaccine in African Infants

The RTS,S Clinical Trials Partnership

ABSTRACT

A Per-Protocol Population



No. at Risk

RTS,S/AS01	3995	3692	3309	2845	1272
Control vaccine	2008	1747	1501	1294	600

Late 2013 results—47% efficacy in children over longer follow up
2019—target rollout in 3 sub-Saharan countries for childhood vaccination

Efficacy and safety of RTS,S/AS01 malaria vaccine with or without a booster dose in infants and children in Africa: final results of a phase 3, individually randomised, controlled trial



RTS,S Clinical Trials Partnership*

www.thelancet.com Published online April 24, 2015 [http://dx.doi.org/10.1016/S0140-6736\(15\)60721-8](http://dx.doi.org/10.1016/S0140-6736(15)60721-8)

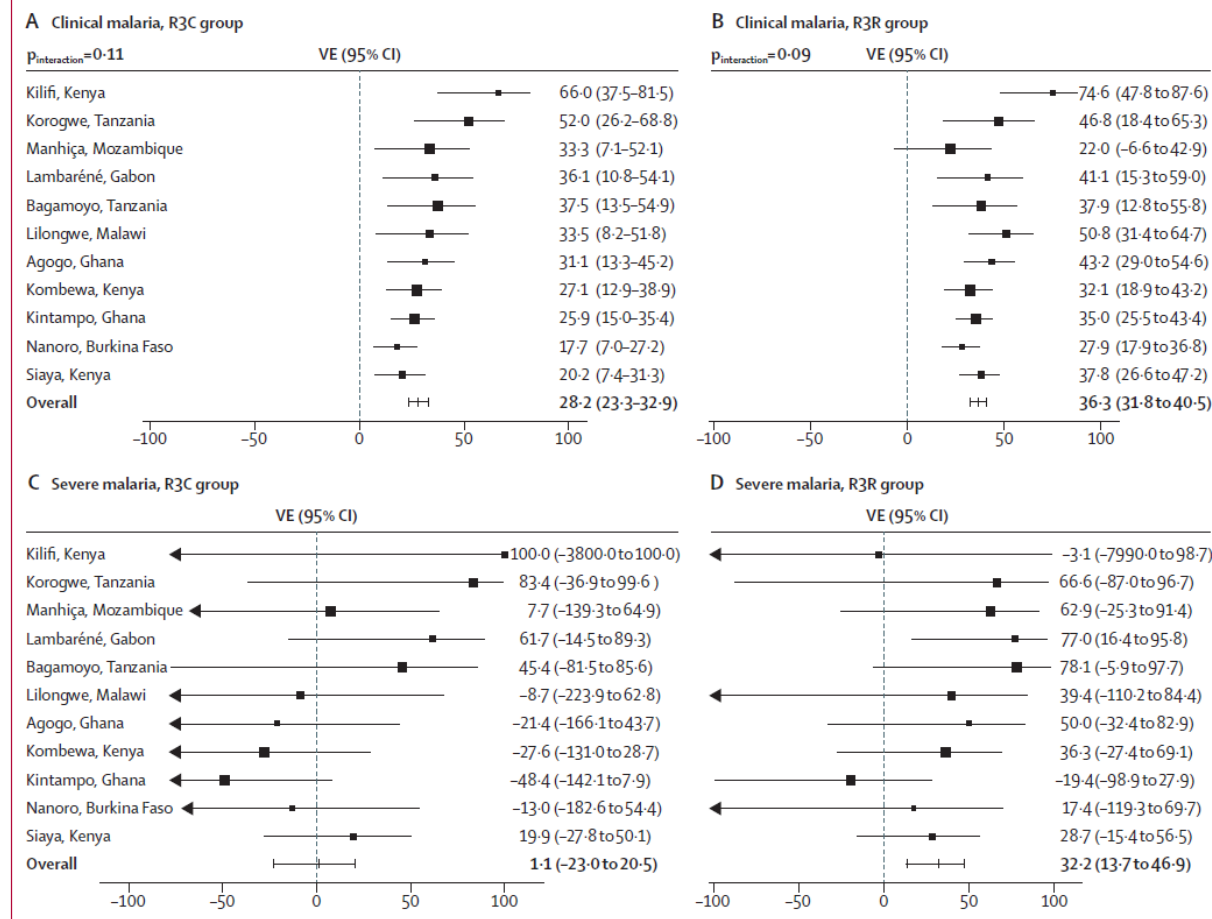


Figure 3: Vaccine efficacy against clinical and severe malaria by study site in the 5-17 months age category

VE against all episodes of clinical malaria (primary case definition) in (A) the R3C group and (B) the R3R group from month 0 to study end; and VE against severe malaria (primary case definition) in (C) the R3C group and (D) the R3R group from month 0 to study end. Study sites are ordered from lowest (Kilifi) to highest (Siaya) incidence of clinical malaria (secondary case definition) measured in control infants 6-12 weeks of age at enrolment during 12 months of follow-up. $p_{\text{interaction}}$ not calculated for (C) or (D). Analyses were by modified intention to treat. Bars are 95% CIs. The size of each square is proportional to the number of participants enrolled at each study site. The following numbers of children aged 5-17 months were enrolled by site for all three groups (R3R, R3C, and C3C) together: 600 in Kilifi, 912 in Korogwe, 1002 in Manhiça, 704 in Lambaréné, 903 in Bagamoyo, 800 in Lilongwe, 600 in Agogo, 1000 in Kombewa, 1002 in Kintampo, 600 in Nanoro, and 799 in Siaya. R3C=RTS,S/AS01 primary schedule without booster. C3C=control group. R3R=RTS,S/AS01 primary schedule with booster. VE=vaccine efficacy.

MATH MODELING AND MALARIA

- Transmission models have contributed substantially to the understanding of malaria control
- Within host modeling is crucial to determine the potential efficacy of the three types of vaccine candidates
- The “threshold” effects of malaria infection (immunity is helpful in endemic regions, but requires frequent low grade re-infection) are particularly suited to a quantitative approach



TRYPANASOMA CRUZI

- Protozoan parasite
- Complex life cycle initiated by triatomine bite
- Long term causative agent of cardiomyopathy
- Limited treatment options and confusion over pathogenesis

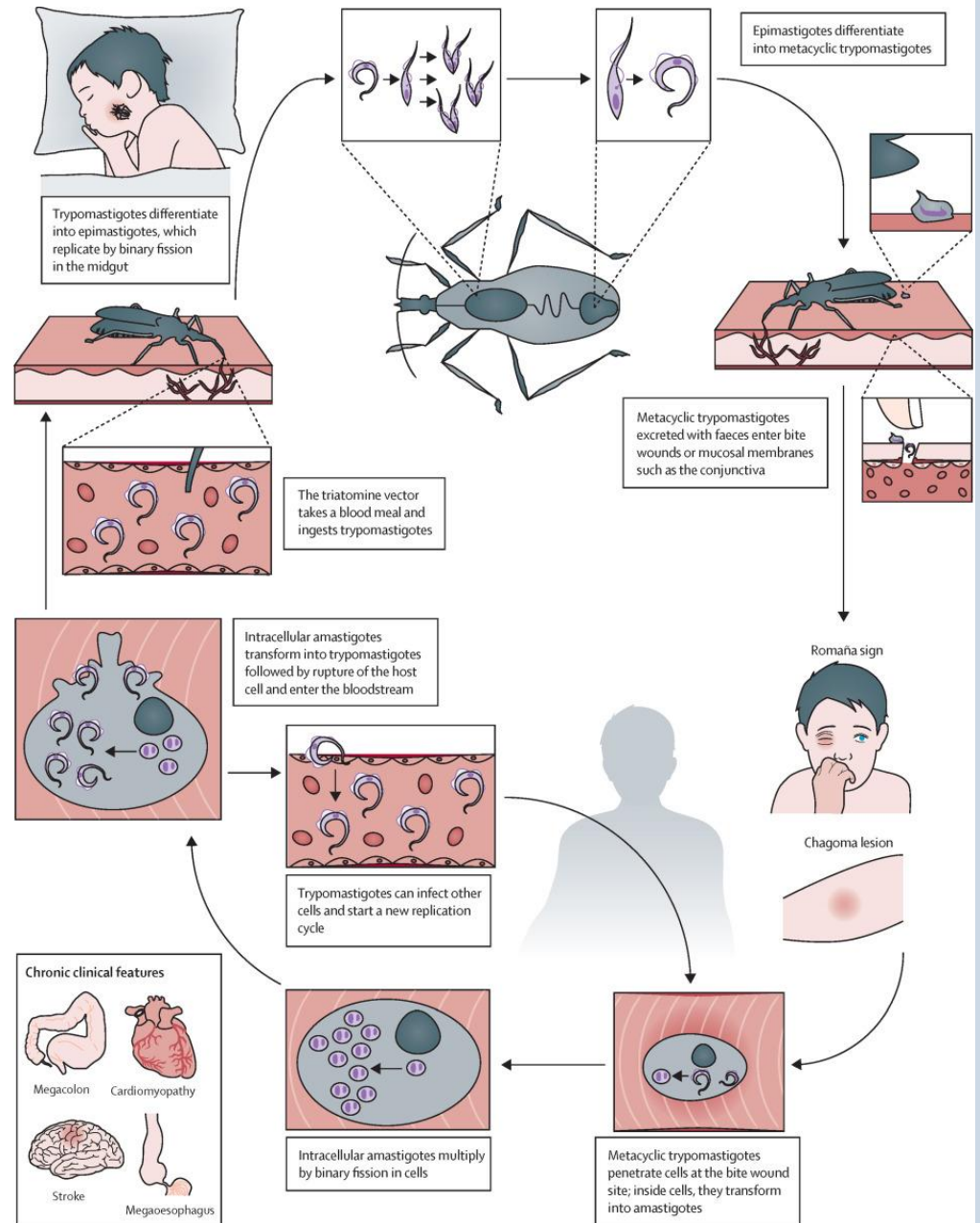
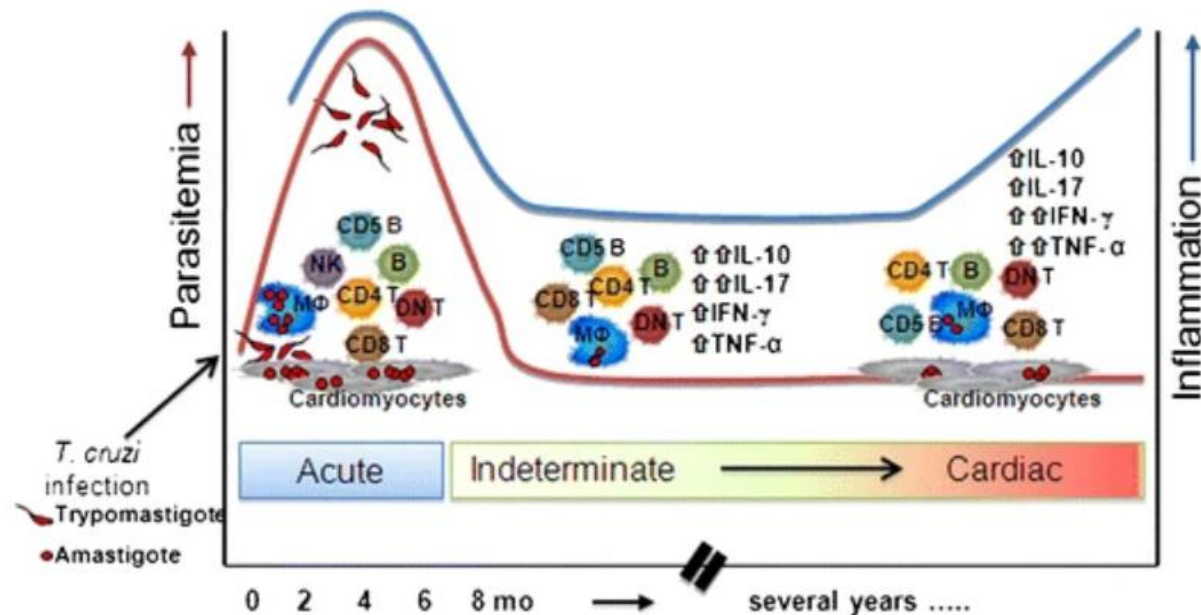


Table 1 Changes in prevalence, incidence, and mortality of Chagas disease, 1985–2010, in 21 endemic countries in Latin America

		1980–85	2005	2010
Population at risk (% total)		92 895 000 (25%)	108 595 000 (20·4%)	70 199 360 (12·9%)
Number of infected people		17 395 000	7 694 500	5 742 167
Number of new cases per year		700 000	55 585	38 593
	Congenital transmission	7000–49 000 *	14 385	8668
	Vectorial transmission	Not reported	41 200	29 925
Number of deaths per year		>45 000	12 500	12 000

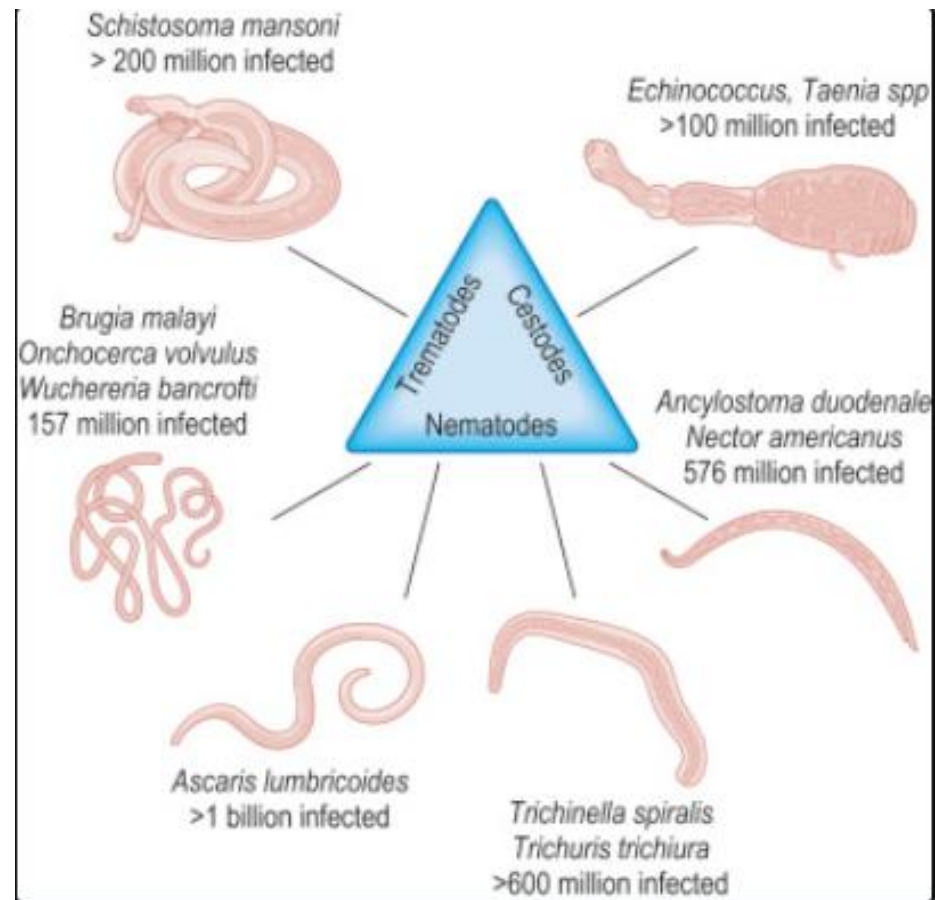
* Estimation based on data provided in the report in reference 10. Data are from references 10, 29, 32, 33.

The pathogenesis of chronic chagasic cardiomyopathy is not completely understood and, until recently, the consequences of the disease were considered to be autoimmune. Although several autoantigens that cross-react with *T. cruzi* antigens and autoantibodies have been identified, the role of autoimmunity in pathogenesis is unknown.⁶³ Recent evidence has shown that tissue damage is the principal role of *T. cruzi* and the chronic inflammatory response that *T. cruzi* elicits.^{64,65} There is a growing consensus that the balance between persistence of infection and host immune response is crucial for the establishment and progression of cardiomyopathy.^{66,67} During the chronic phase, inflammation is the main determinant of progression; the other potential contributory factors are virulence of the *T. cruzi* strain and tissue tropism.^{66–68} An inflammatory environment predominates in the cardiac form, with the production of cytokines such as interferon γ and tumour necrosis factor α and other cytotoxic mechanisms involving CD8+ T cells, which lead to tissue damage and, ultimately, to severe cardiomyopathy; in the indeterminate clinical form, a regulatory immune response (characterised by production of interleukin 10 and interleukin 17)



THE BURDEN OF HELMINTH INFECTIONS

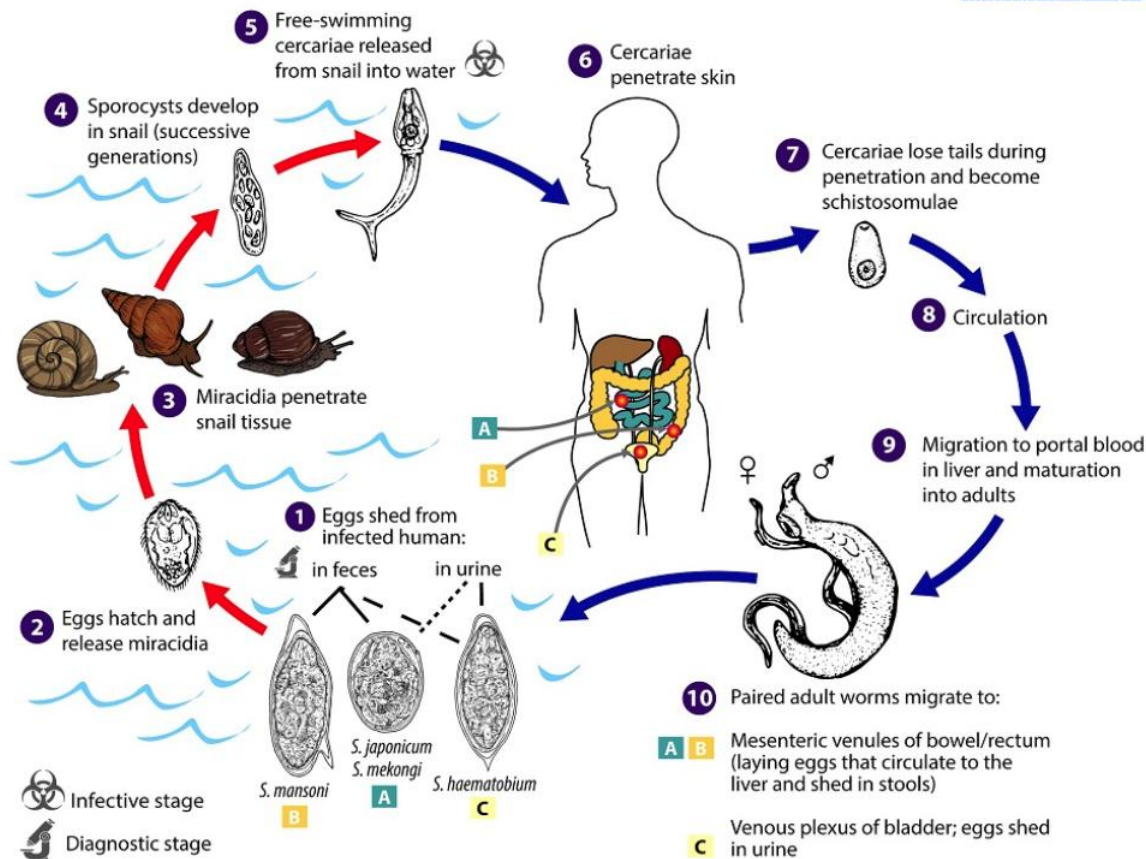
- After malaria, helminths represent the biggest DALY-impact of a parasitic infection in human populations
- Multiple life cycles drive distinct pathological outcomes



SCHISTOSOMA LIFE CYCLE

1DPDx

Schistosoma spp.

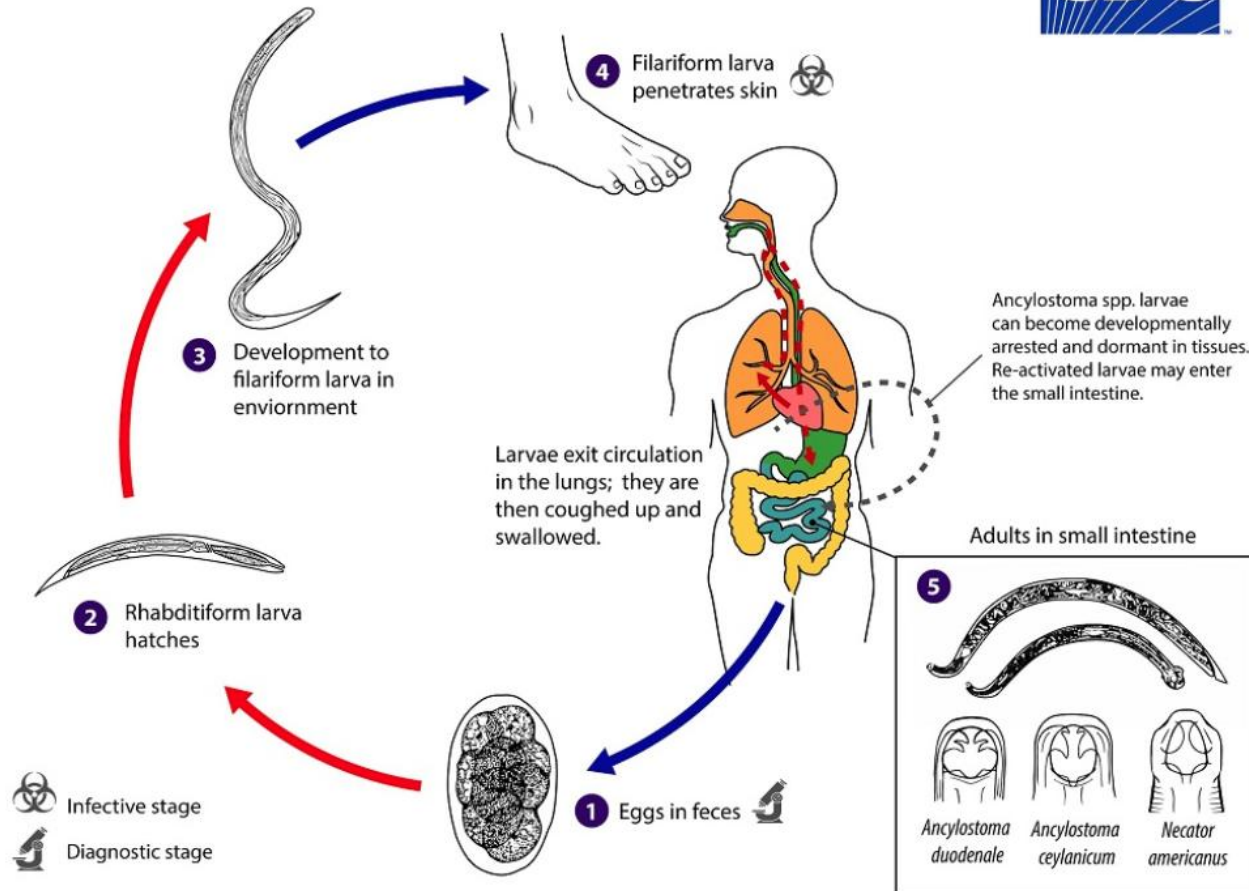


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HOOKWORM LIFE CYCLES

4DPDx

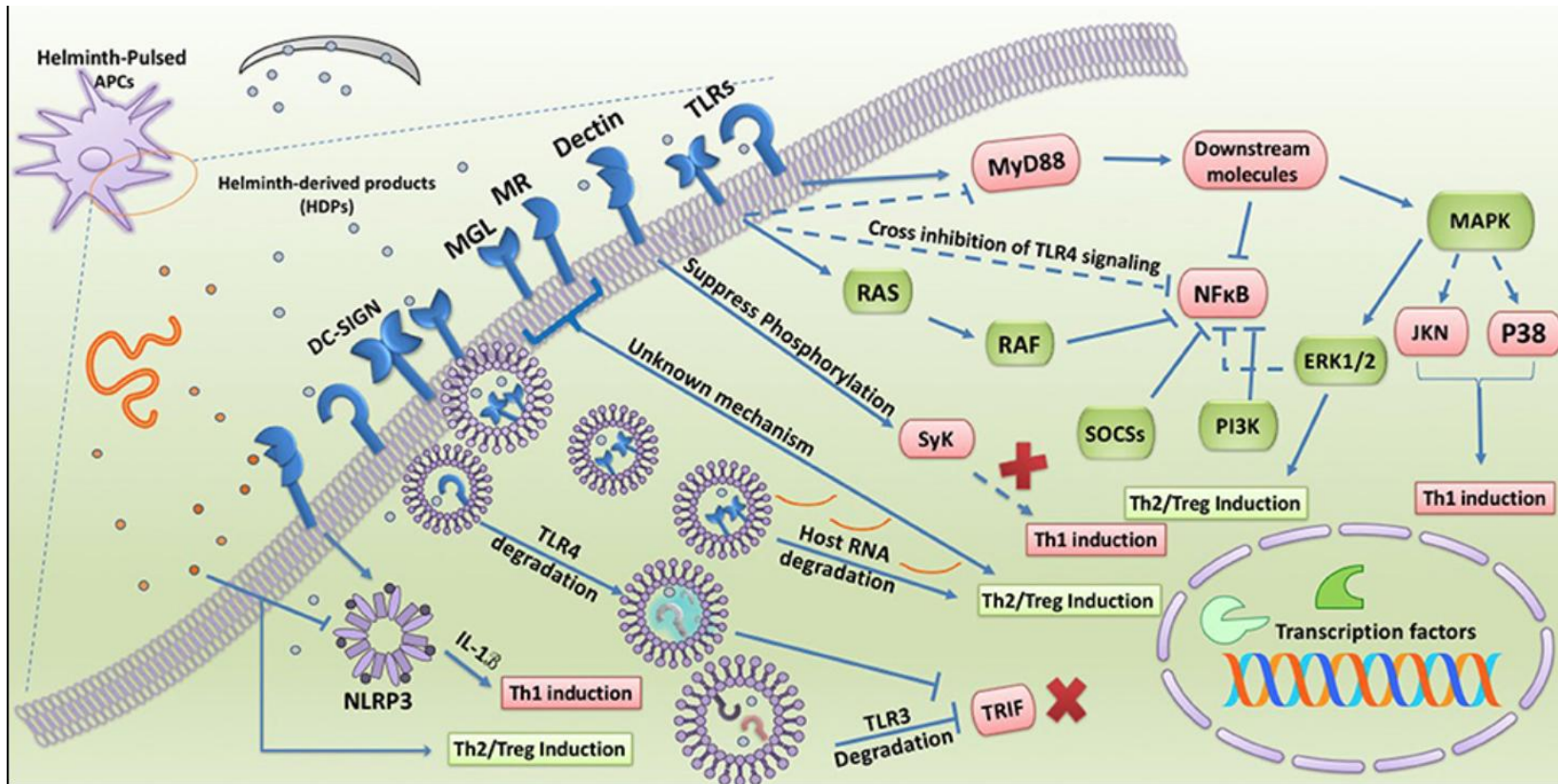
Intestinal Hookworm

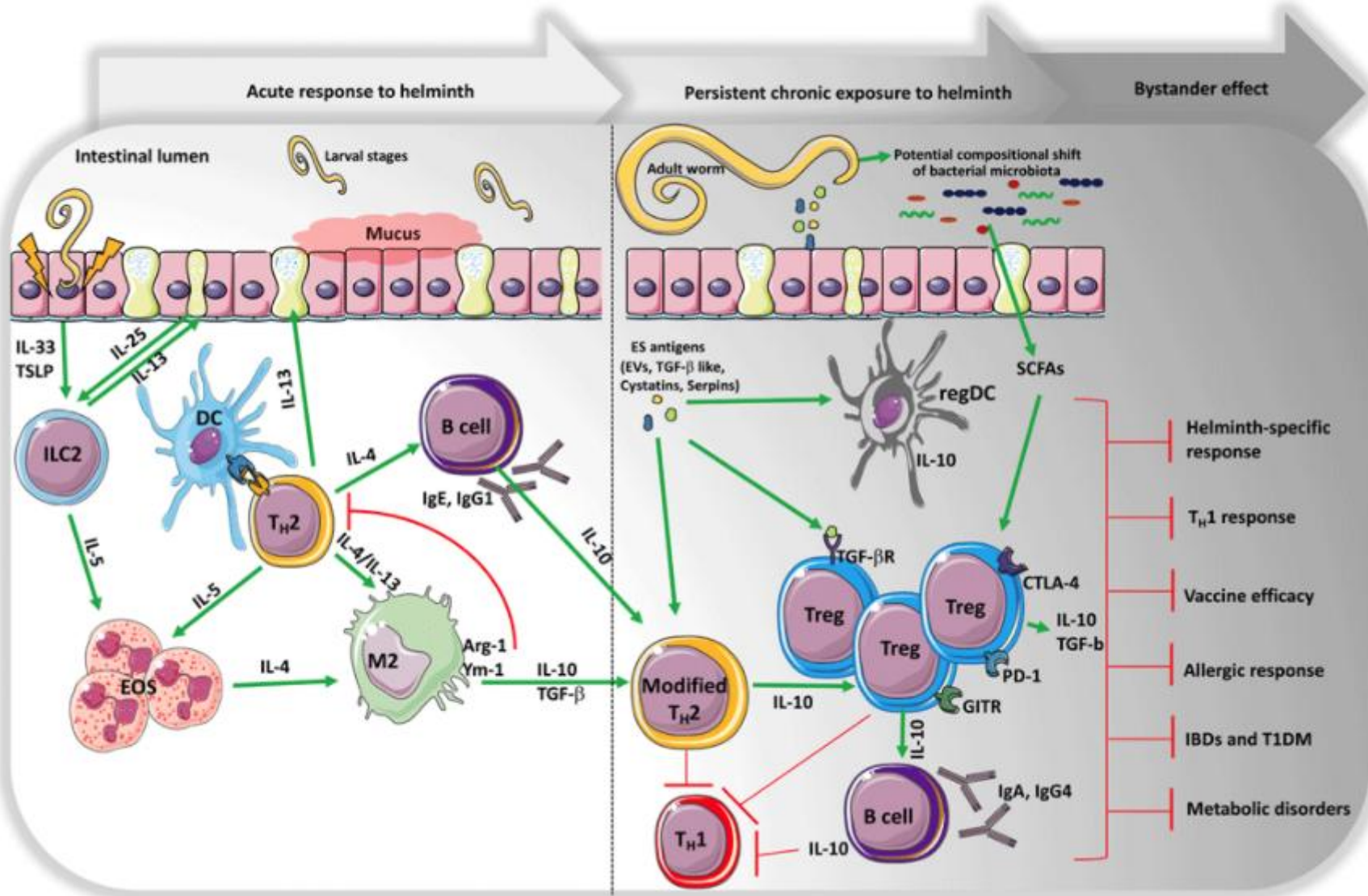


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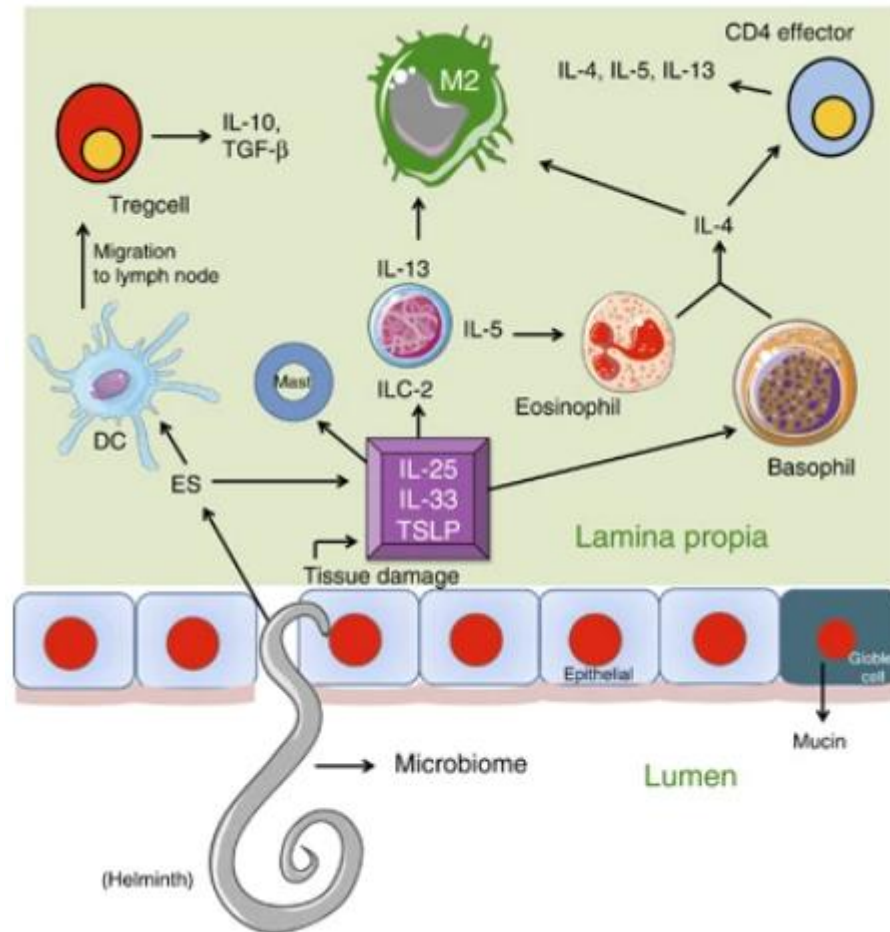


INNATE SIGNALING IN RESPONSE TO HELMINTH-DERIVED PRODUCTS





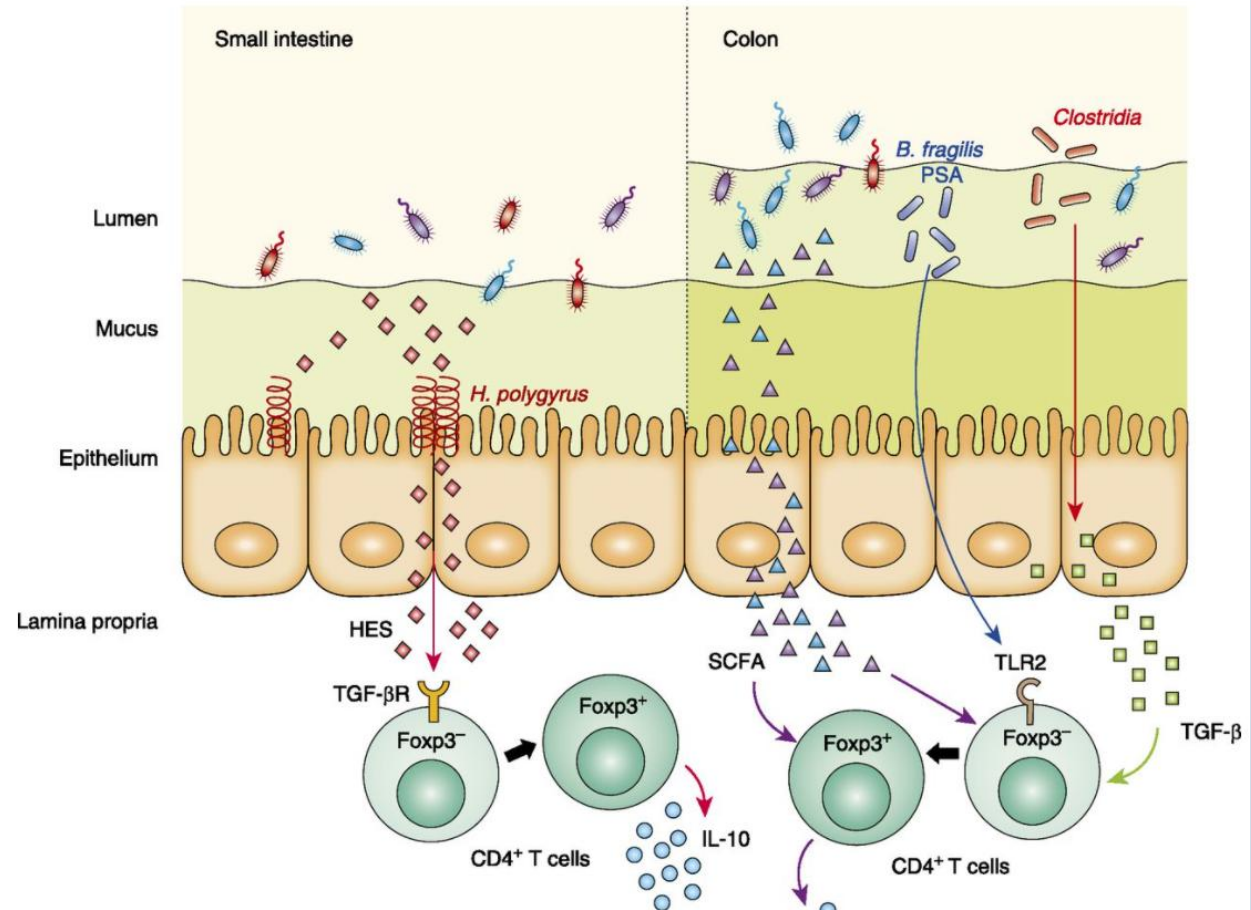
LOCAL TISSUE INDUCTION OF REGULATORY RESPONSES



- Mucosal Immunology volume 7, pages 53–702 (2014)

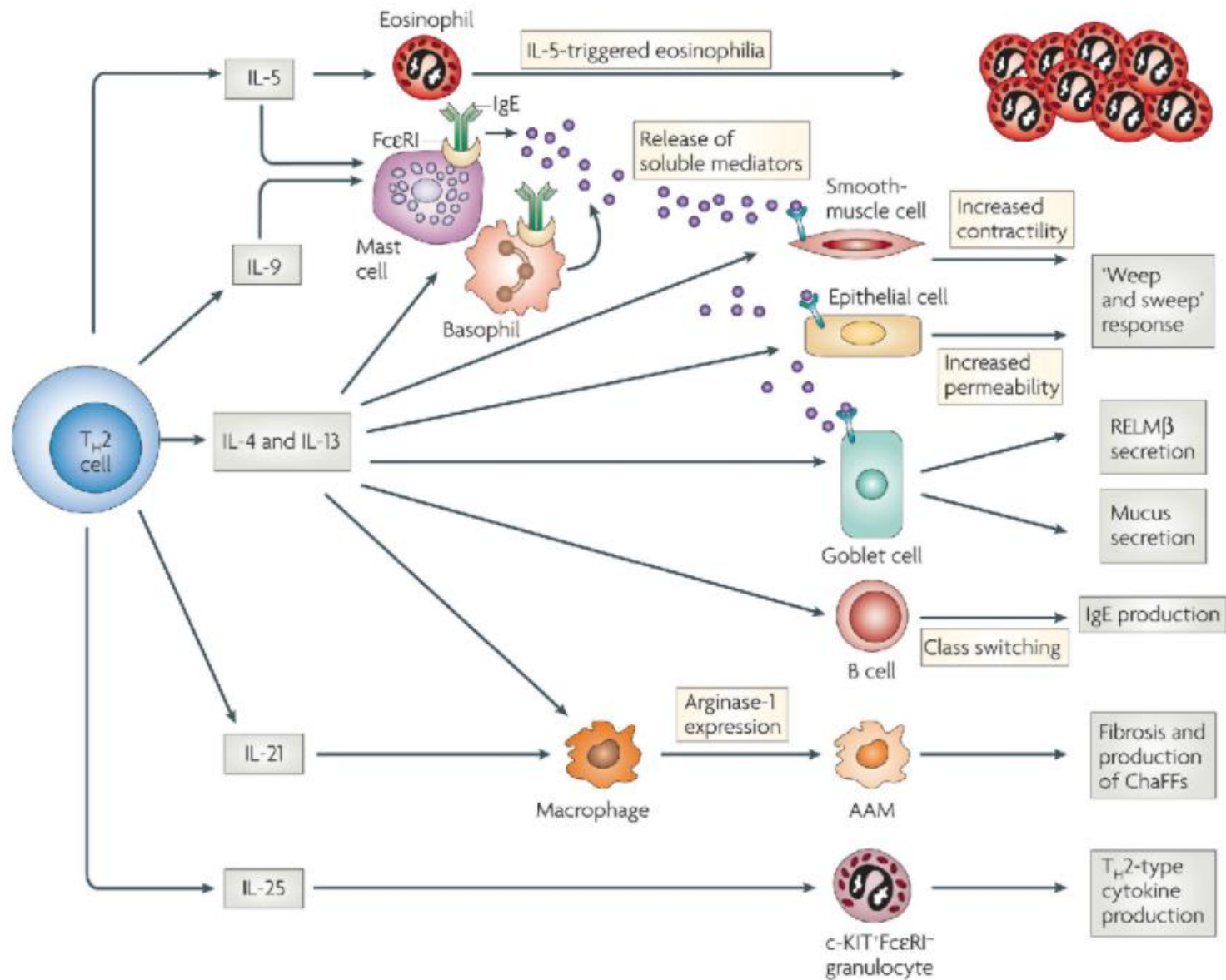
HELMINTH INTERACTION WITH THE MICROBIOME

- Microbiome components and helminths likely co-evolved
- Similar effector and host manipulation pathways
- Immune tolerance vs. immune resistance



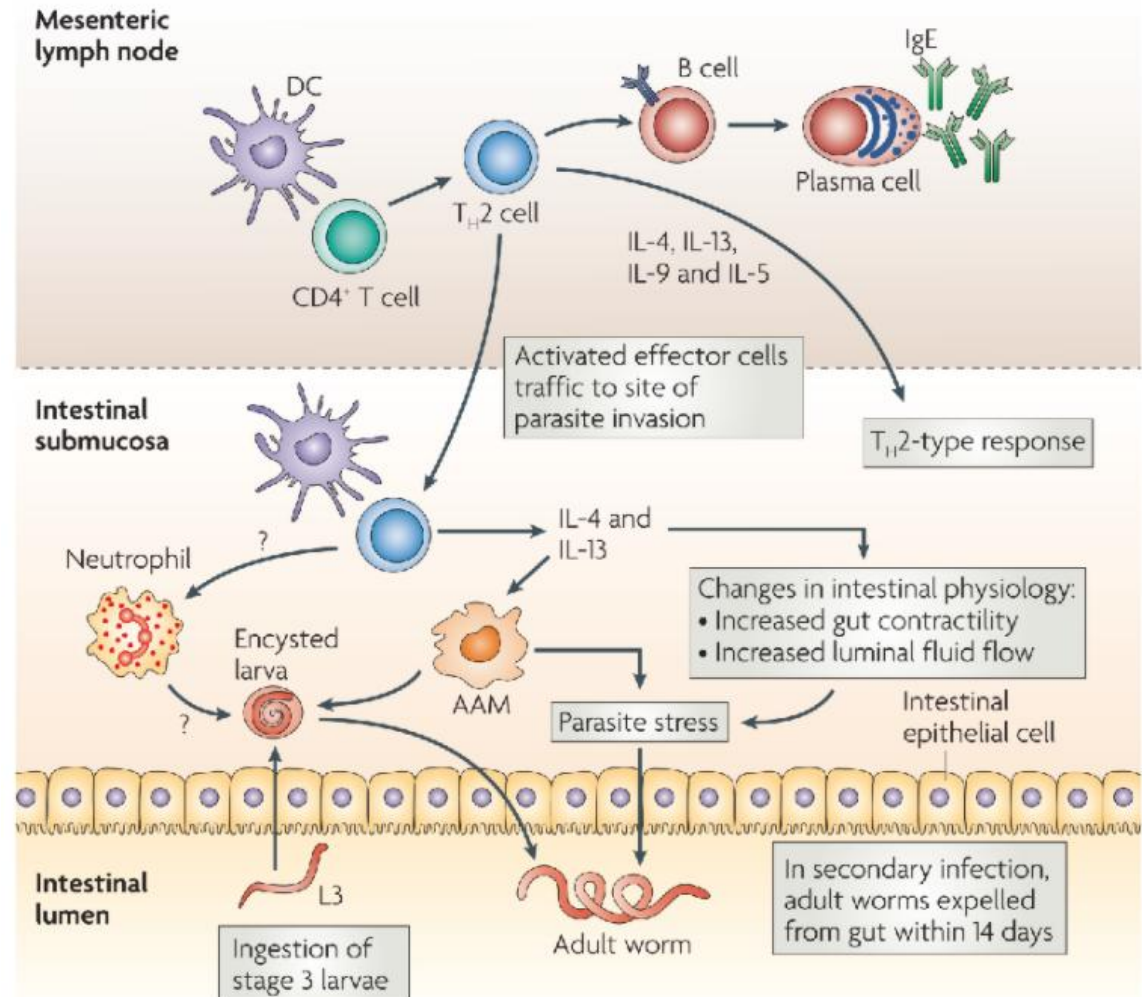
J Immunol November 1,
2015, 195 (9) 4059-4066; DOI:
<https://doi.org/10.4049/jimmunol.15014>

THE TH2 RESPONSE IS THE MAIN EFFECTOR OF HELMINTH CLEARANCE

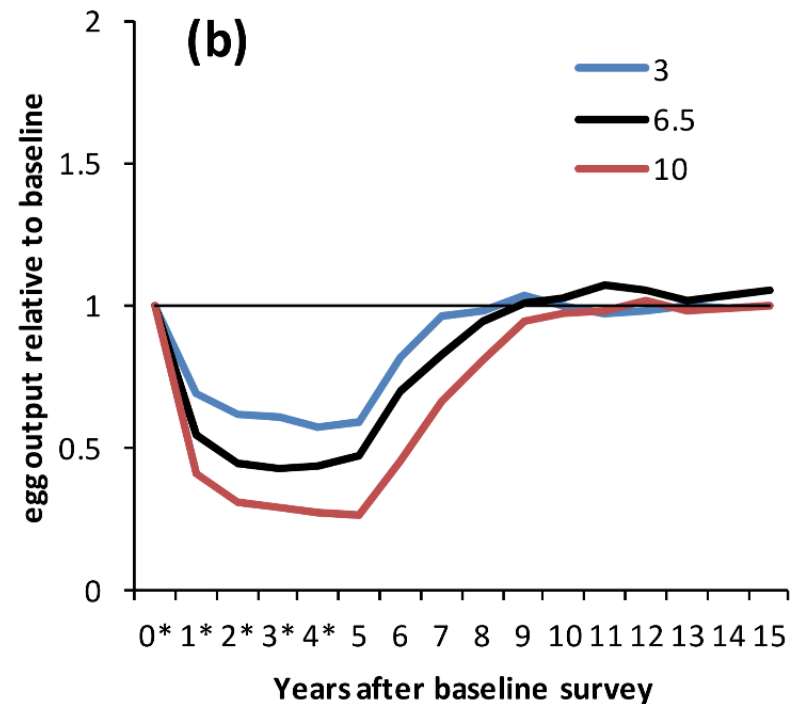
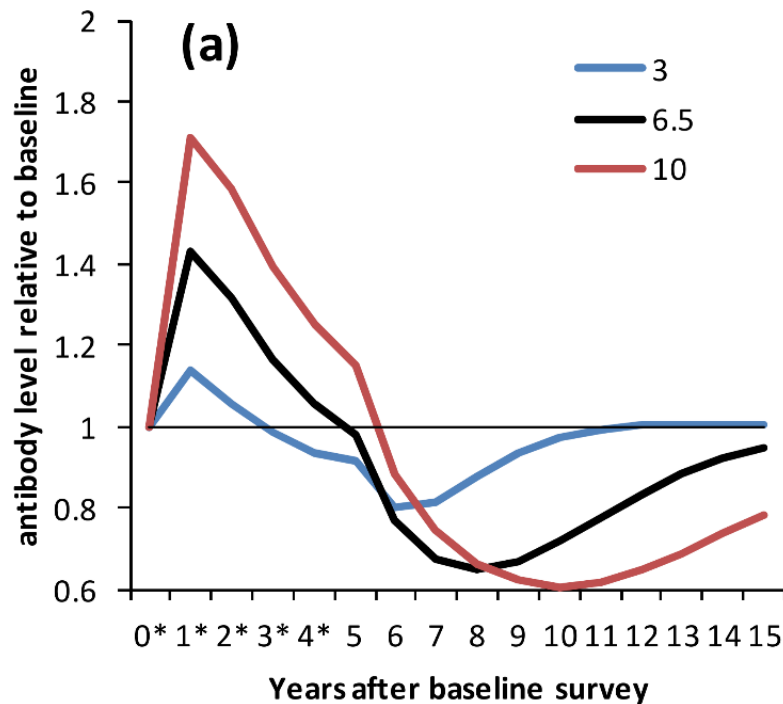


EFFECTOR MECHANISMS FOR HELMINTH DAMAGE AND CLEARANCE

- CD4 T cell orchestrate the response against helminths
- Th2 responses involve multiple effector mechanisms, not all fully understood
- Cell based and mechanical disruptions lead to worm death and expulsion



SOME EVIDENCE FOR PROTECTIVE IMMUNITY

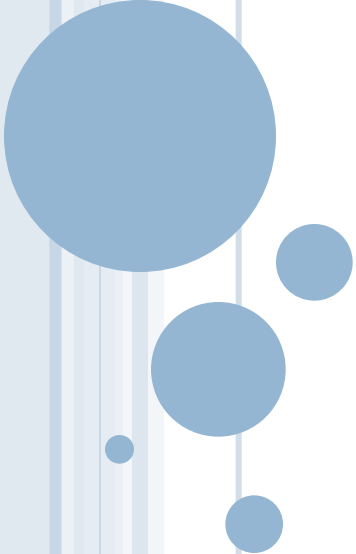


- IgE levels correlate with egg burdens and rates of re-infection
- <https://doi.org/10.1371/journal.pntd.0003059>

CONCLUSIONS/DISCUSSION POINTS

- Parasitic helminths represent a major evolutionary burden on human development
- An entire arm of the immune response is dedicated to response to worm infections, but in many populations it fails to be engaged
- Evolutionary “assumption” may have led to increases of atopic conditions, such as allergy, resulting from lack of helminth targets





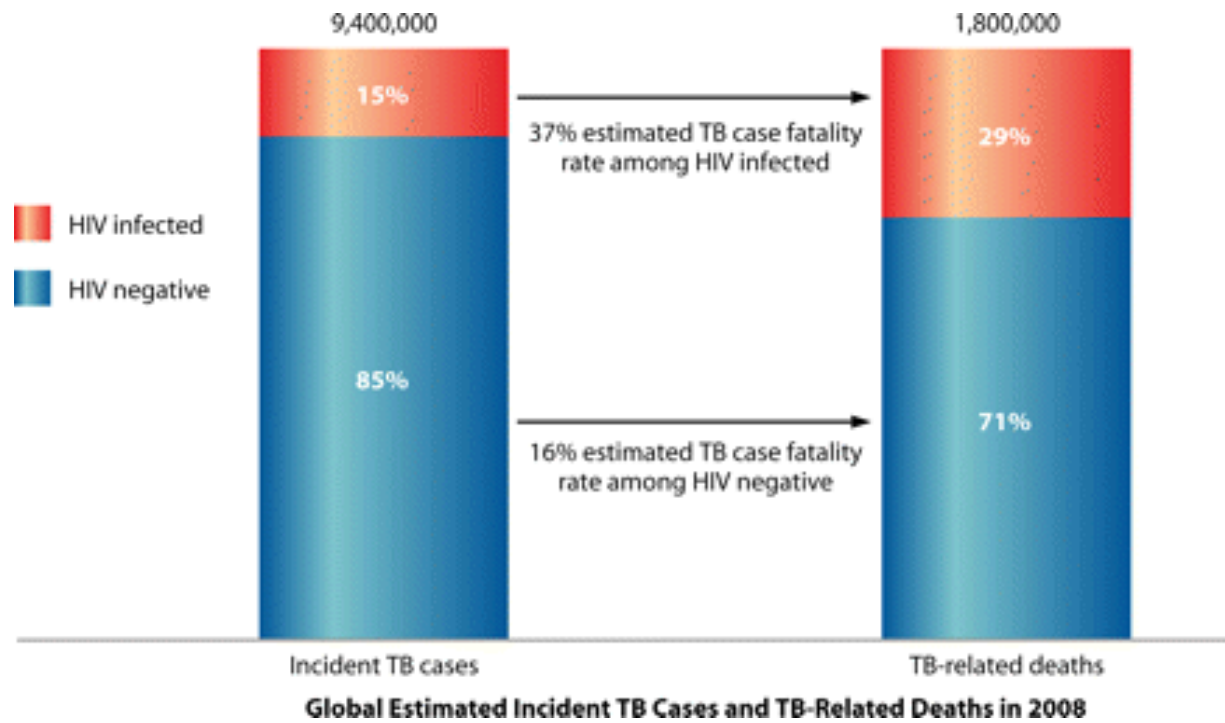
IMMUNOLOGICAL CONSIDERATIONS IN CO-INFECTIONS

HIV

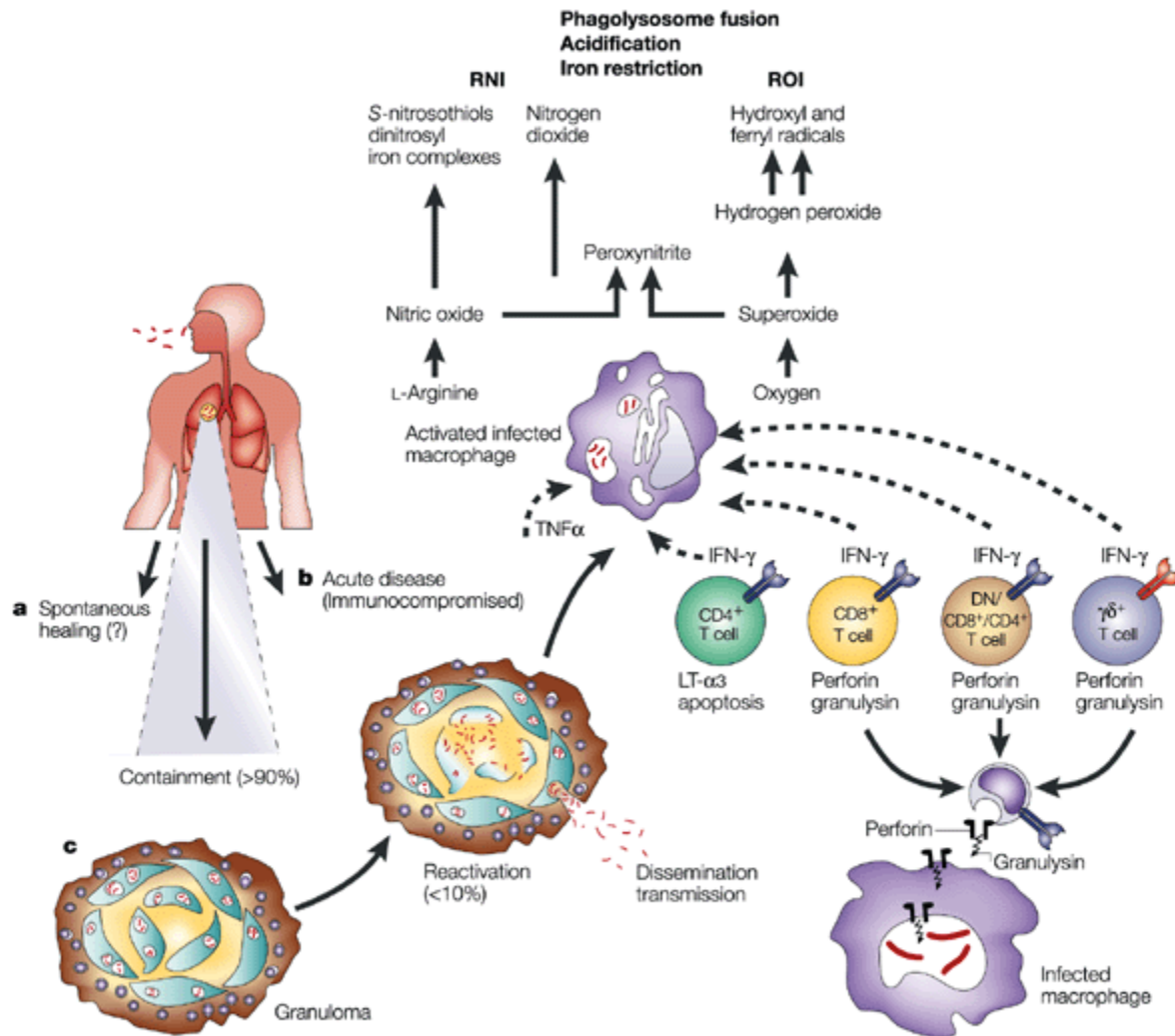
- Two directions of interaction
 - HIV on other infections
 - Other infections on HIV
- What are the underlying mechanisms of each?
 - Immune dysregulation, no CD4 cells
 - Systemic bias away from protective immunity and viral control



HIV AND TB



CONTROL OF MTB



TREATMENT OF HIV PREVENTS TB

	HAART			Non-HAART			Adjusted risk ratio (95%CI)	p	Adjusted number of cases averted (95% CI)
	Number of cases of tuberculosis	Patient- years	Incidence*	Number of cases of tuberculosis	Patient- years	Incidence*			
Overall	9	375.1	2.4	82	848.2	9.7	0.19 (0.09–0.38)	<0.0001	7.3 (4.7–9.8)
CD4 count (cells/μL)									
<200	5	148	3.4	41	235	17.5	0.18 (0.07–0.47)	<0.0001	14.2 (9.7–19.7)
200–350	2	121.2	1.7	27	225	12.0	0.12 (0.03–0.53)	<0.0001	10.6 (6.8–15.9)
>350	2	100.1	2.0	14	388.3	3.6	0.36 (0.1–1.74)	0.78	2.3 (1.1–4.4)
WHO stage									
1 or 2	1	219	0.5	36	657.4	5.5	0.08 (0.01–0.57)	0.01	5.1 (3.45–7.1)
3 or 4	8	172.75	4.6	46	190.8	24.1	0.22 (0.09–0.41)	0.03	18.8 (13.2–26.1)
Socioeconomic status									
Low	6	166.21	3.6	65	514.34	10.9	0.21 (0.09–0.49)	<0.0001	8.6 (6.2–11.5)
High	3	208.89	1.44	17	333.86	5.09	0.17 (0.05–0.57)	<0.0001	4.2 (2.3–7.0)

HAART=highly active antiretroviral therapy. *Per 100 patient-years.

- Disease often manifests in non-traditional ways—extra pulmonary (not necessarily a breakdown of the lung granuloma?)

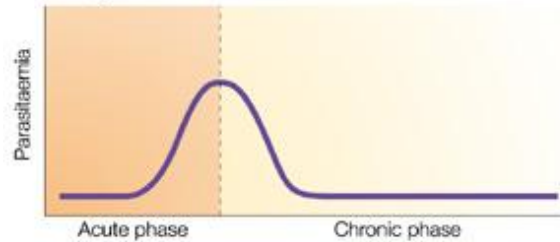
HIV AND MALARIA

- Several studies report increased presentation in children and adults of severe malaria correlating with HIV co-infection
- These studies are not particularly robust or consistent—seems like there *should* be an association
- Studies difficult to do and loaded with confounding variables

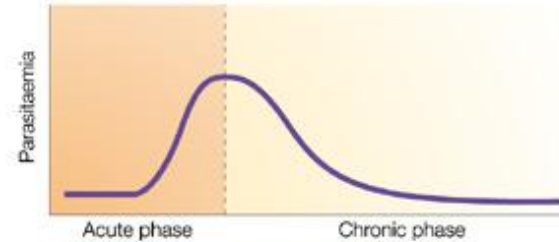


MALARIA IMMUNOLOGICAL CONTROL

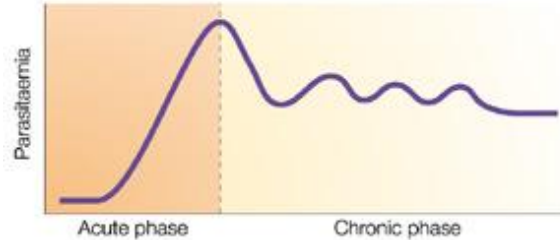
a Wild-type mice



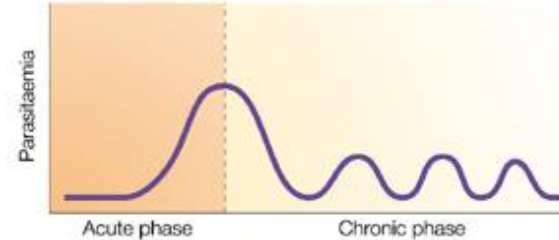
d $\gamma\delta$ T-cell-deficient mice



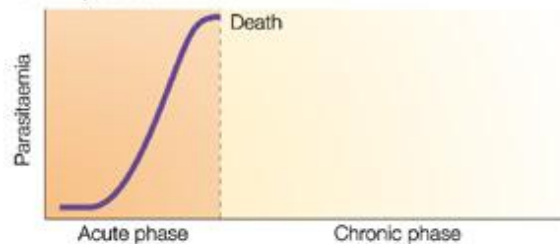
b CD4⁺ T-cell-depleted mice



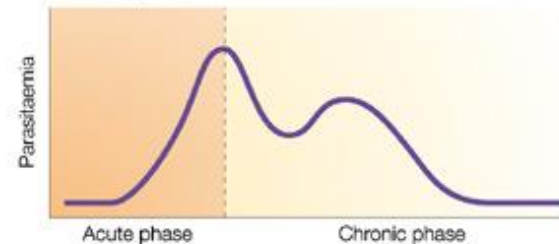
e B-cell-depleted or -deficient mice



c IFN- γ -deficient mice



f NK-cell-depleted mice



WHAT ABOUT THE OTHER DIRECTION?

- Mechanistically, more clear how malaria could increase HIV production
- Robust epidemiological data suggesting greater transmission (viral load? susceptibility?) in malaria endemic areas



HELMINTH CO-INFECTIONS

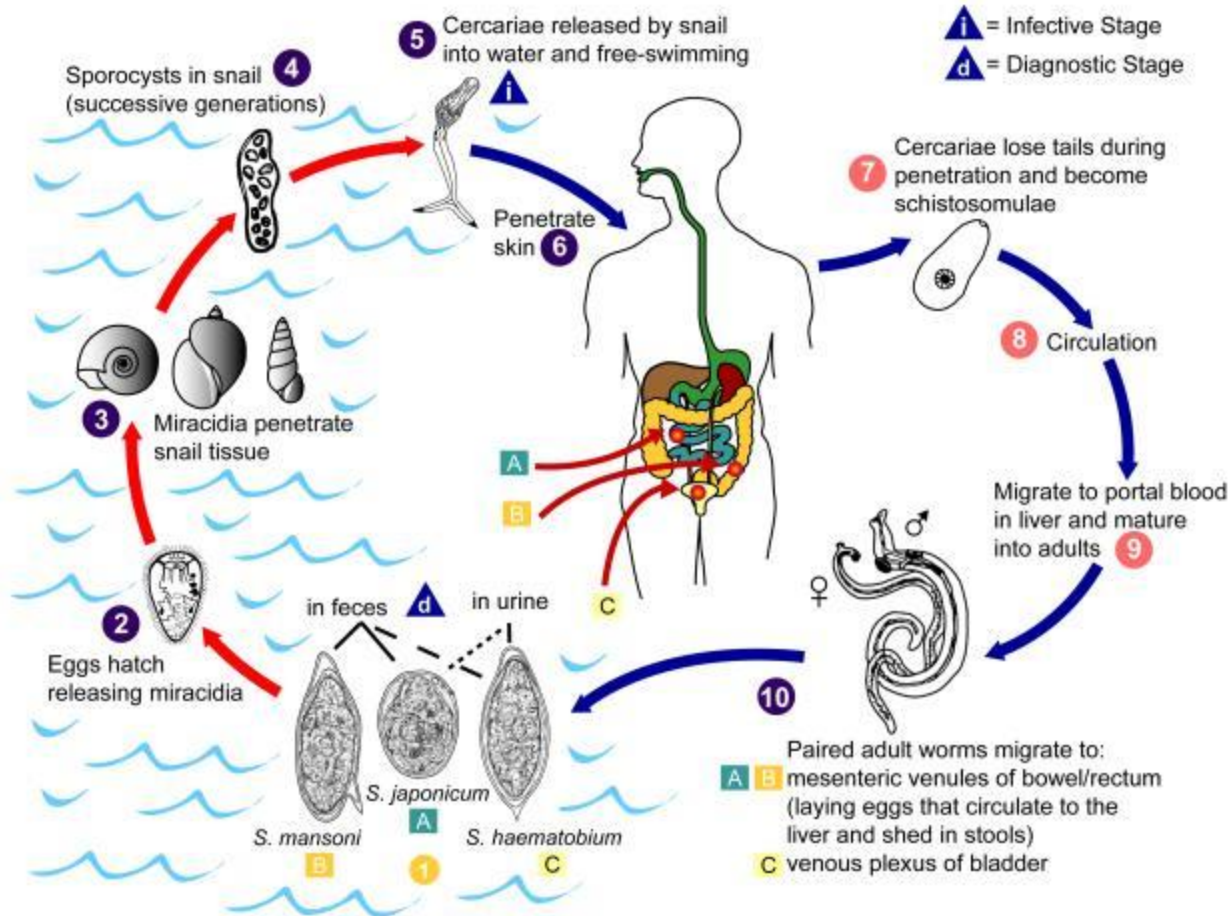
Table 1

The major human helminthiases and their global prevalence and distribution

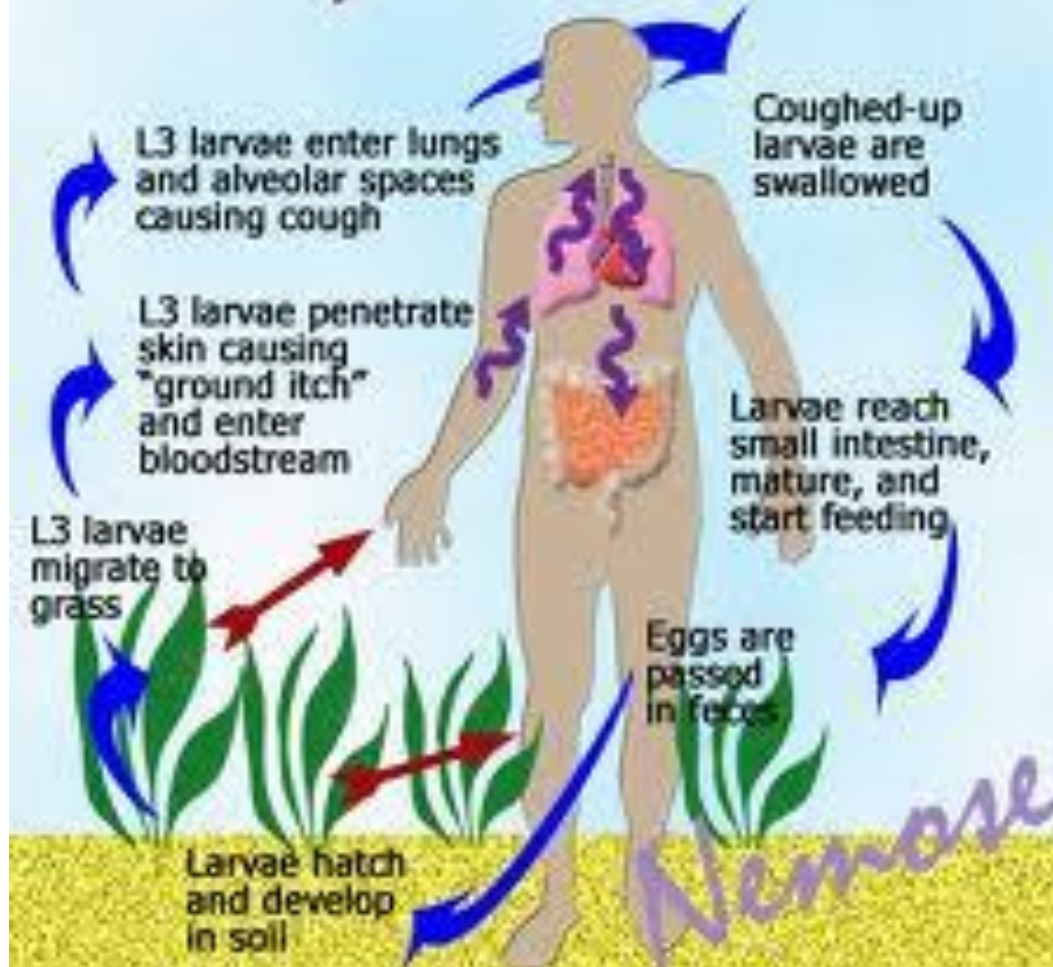
Disease	Major etiologic agent	Global prevalence	Regions of highest prevalence
Soil-transmitted nematodes			
Ascariasis	<i>Ascaris lumbricoides</i> (roundworm)	807 million	Developing regions of Asia, Africa, and Latin America
Trichuriasis	<i>Trichuris trichiura</i> (whipworm)	604 million	Developing regions of Asia, Africa, and Latin America
Hookworm	<i>Necator americanus</i> ; <i>Ancylostoma duodenale</i>	576 million	Developing regions of Asia, Africa, and Latin America (especially areas of rural poverty)
Strongyloidiasis	<i>Strongyloides stercoralis</i> (thread worm)	30–100 million	Developing regions of Asia, Africa, and Latin America (especially areas of rural poverty)
Filarial nematodes			
LF	<i>Wuchereria bancrofti</i> ; <i>Brugia malayi</i>	120 million	Developing regions of India, Southeast Asia, and sub-Saharan Africa
Onchocerciasis (river blindness)	<i>Onchocerca volvulus</i>	37 million	Sub-Saharan Africa
Loiasis	<i>Loa loa</i>	13 million	Sub-Saharan Africa
Dracunculiasis (guinea worm)	<i>Dracunculus medinensis</i>	0.01 million	Sub-Saharan Africa
Platyhelminth flukes			
Schistosomiasis	<i>Schistosoma haematobium</i> ; <i>Schistosoma mansoni</i> ; <i>Schistosoma japonicum</i> (blood flukes)	207 million	Sub-Saharan Africa Sub-Saharan Africa and Eastern Brazil China and Southeast Asia
Food-borne trematodiasis	<i>Clonorchis sinensis</i> (liver fluke); <i>Opisthorchis viverrini</i> (liver fluke); <i>Paragonimus spp.</i> (lung flukes); <i>Fasciolopsis buski</i> (intestinal fluke); <i>Fasciola hepatica</i> (intestinal fluke)	>40 million	Developing regions of East Asia
Platyhelminth tapeworms			
Cysticercosis	<i>Taenia solium</i> (pork tapeworm)	0.4 million (Latin America only)	Developing regions of Asia, Latin America, and sub-Saharan Africa



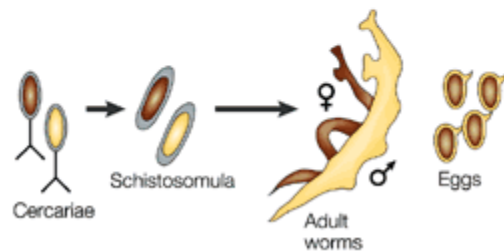
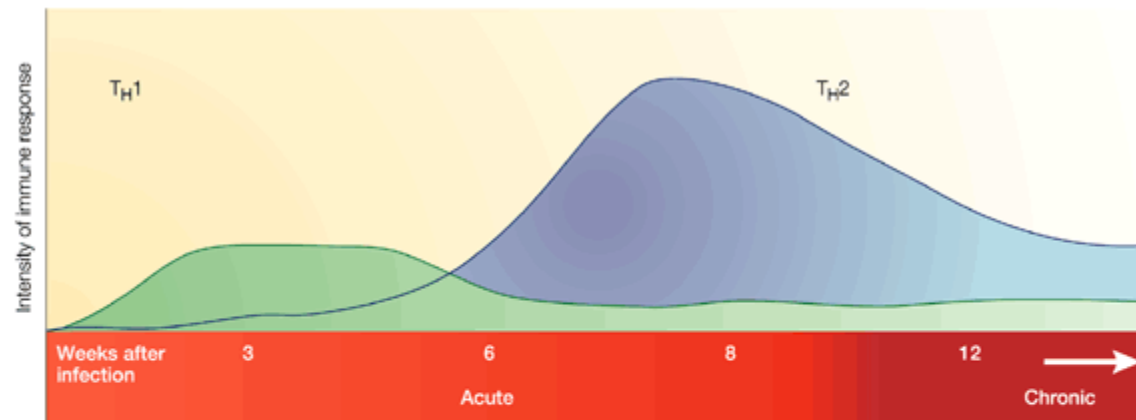
Schistosomiasis

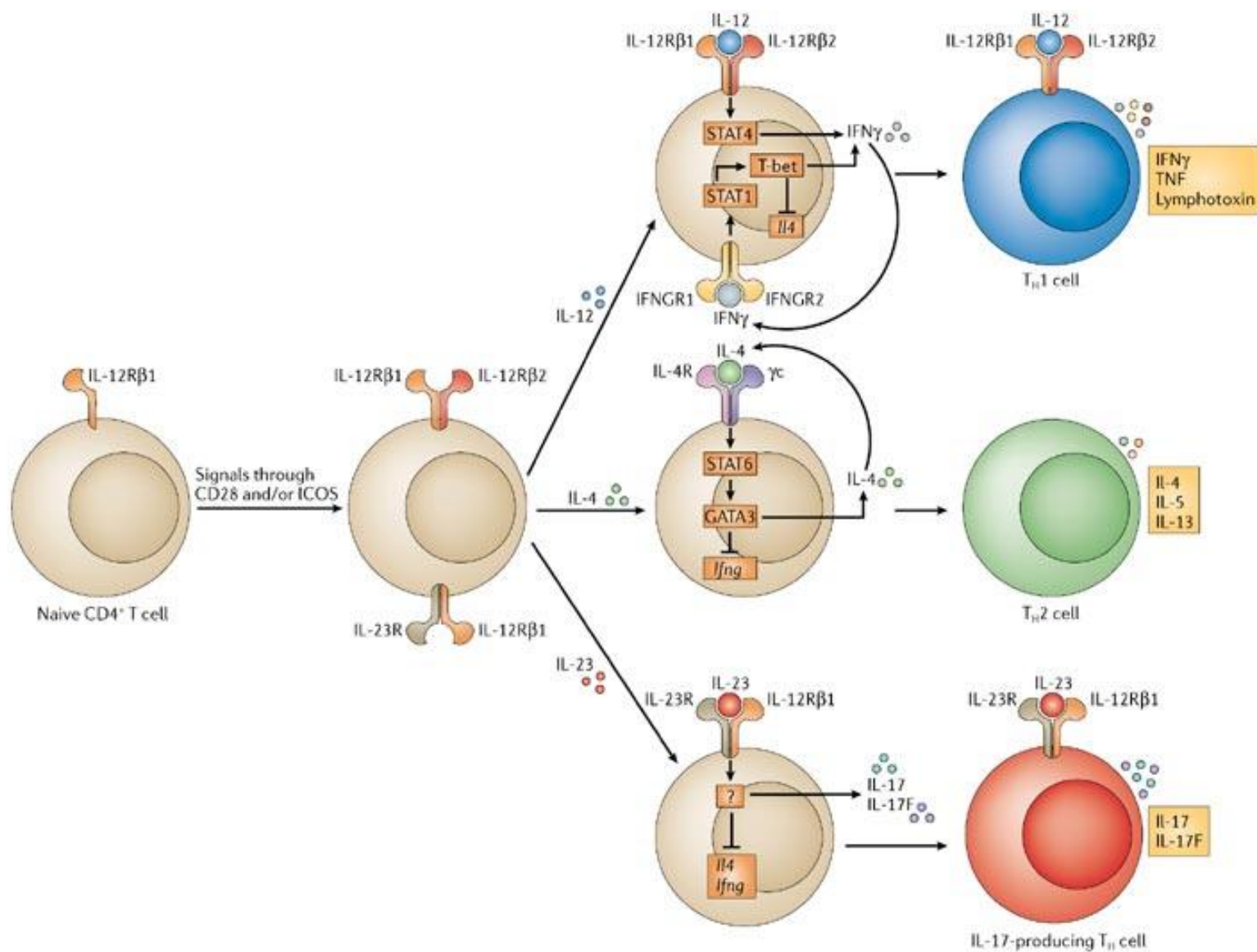


Life cycle of hookworms

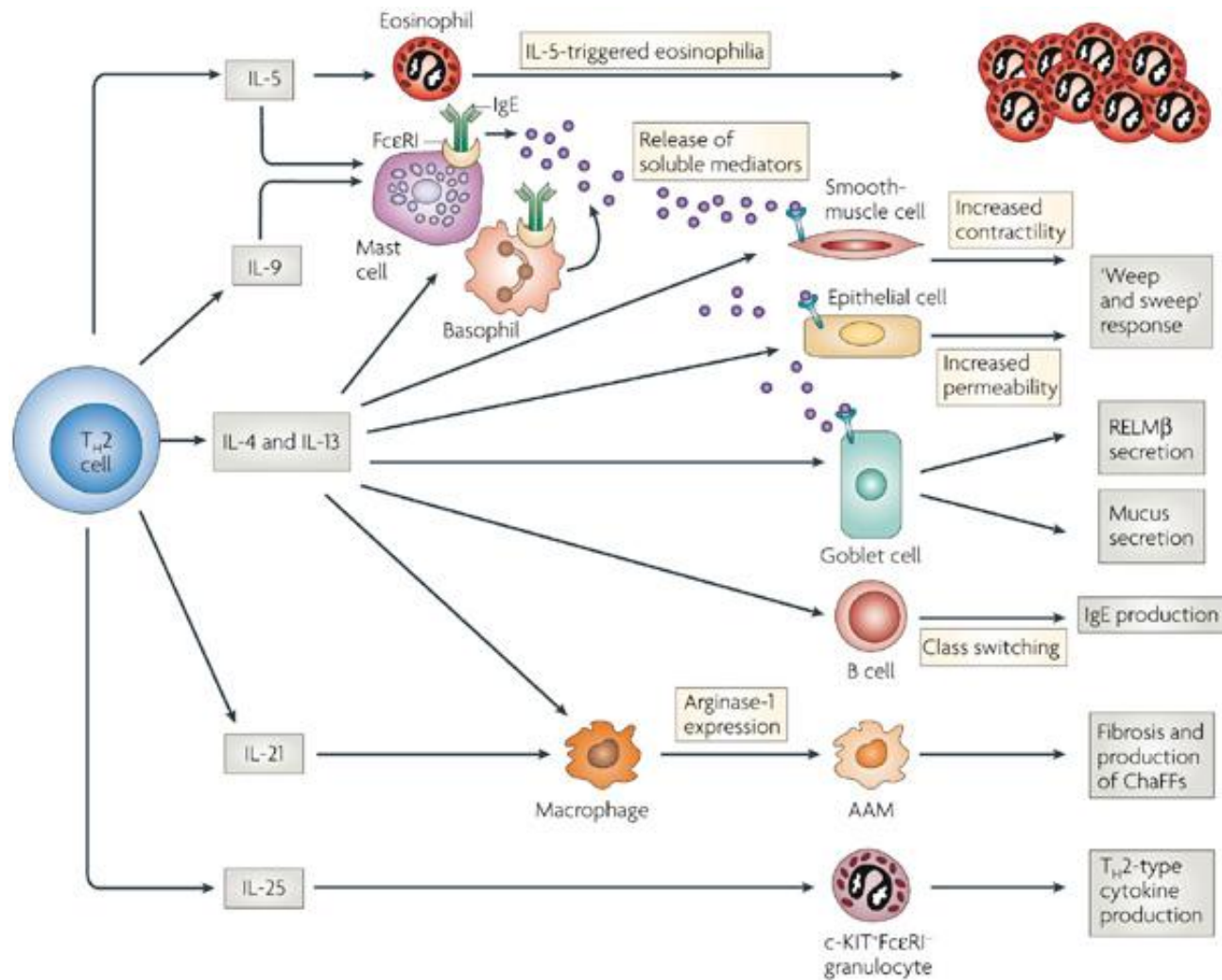


PROGRESSION OF SCHISTOSOME DISEASE





PROTECTIVE IMMUNITY TO HELMINTHS



HOW DO WE ASSESS HIV'S EFFECTS ON WORMS?

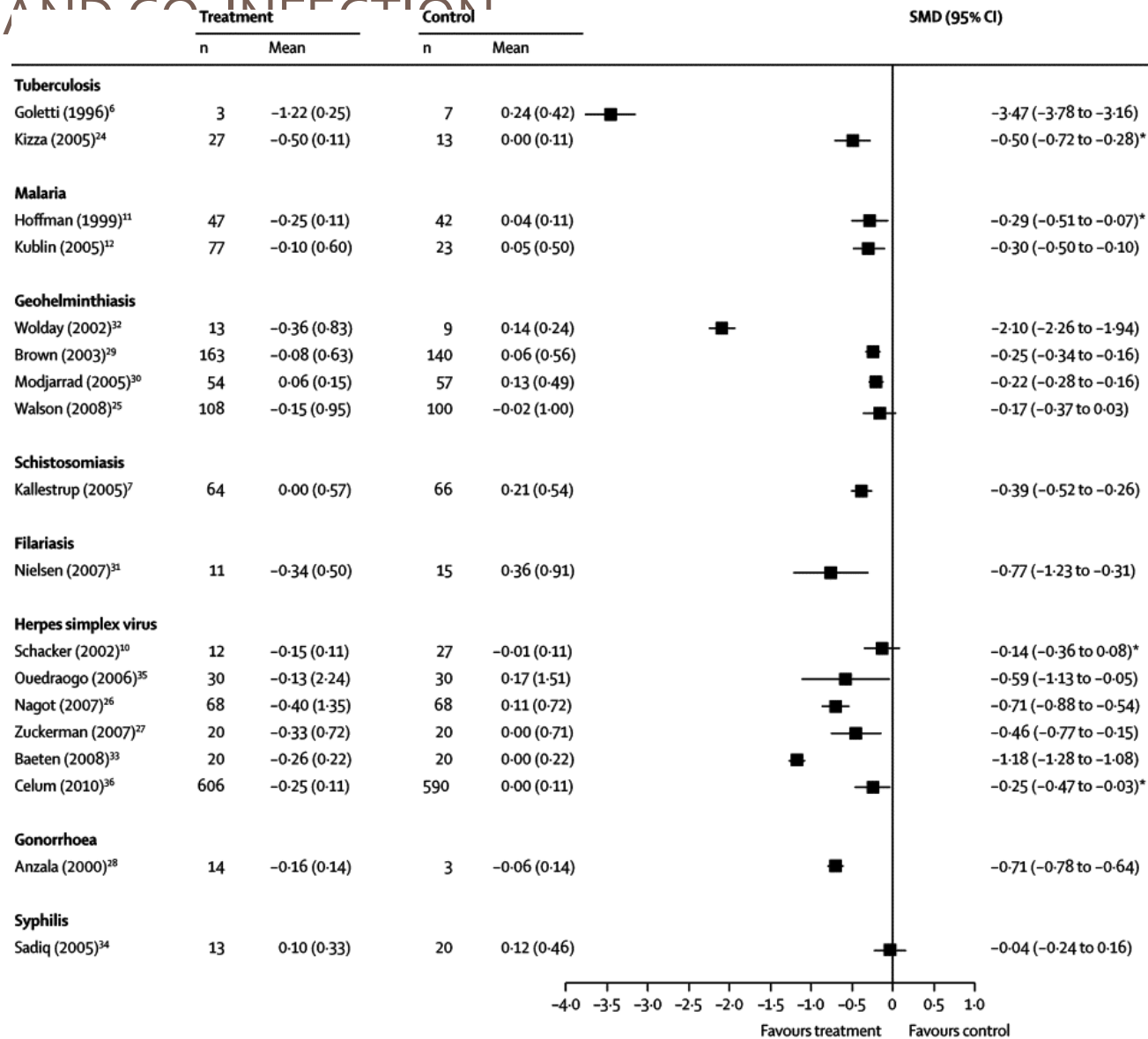
- The natural process of worm clearance is slow and difficult to observe epidemiologically
- Worm burden/egg burden can be measured in endemic areas
- Often, treatment studies are done to measure HIV's relationship to drug efficacy
 - Drugs require antibody/immune competence
- Overall, very limited effects observed



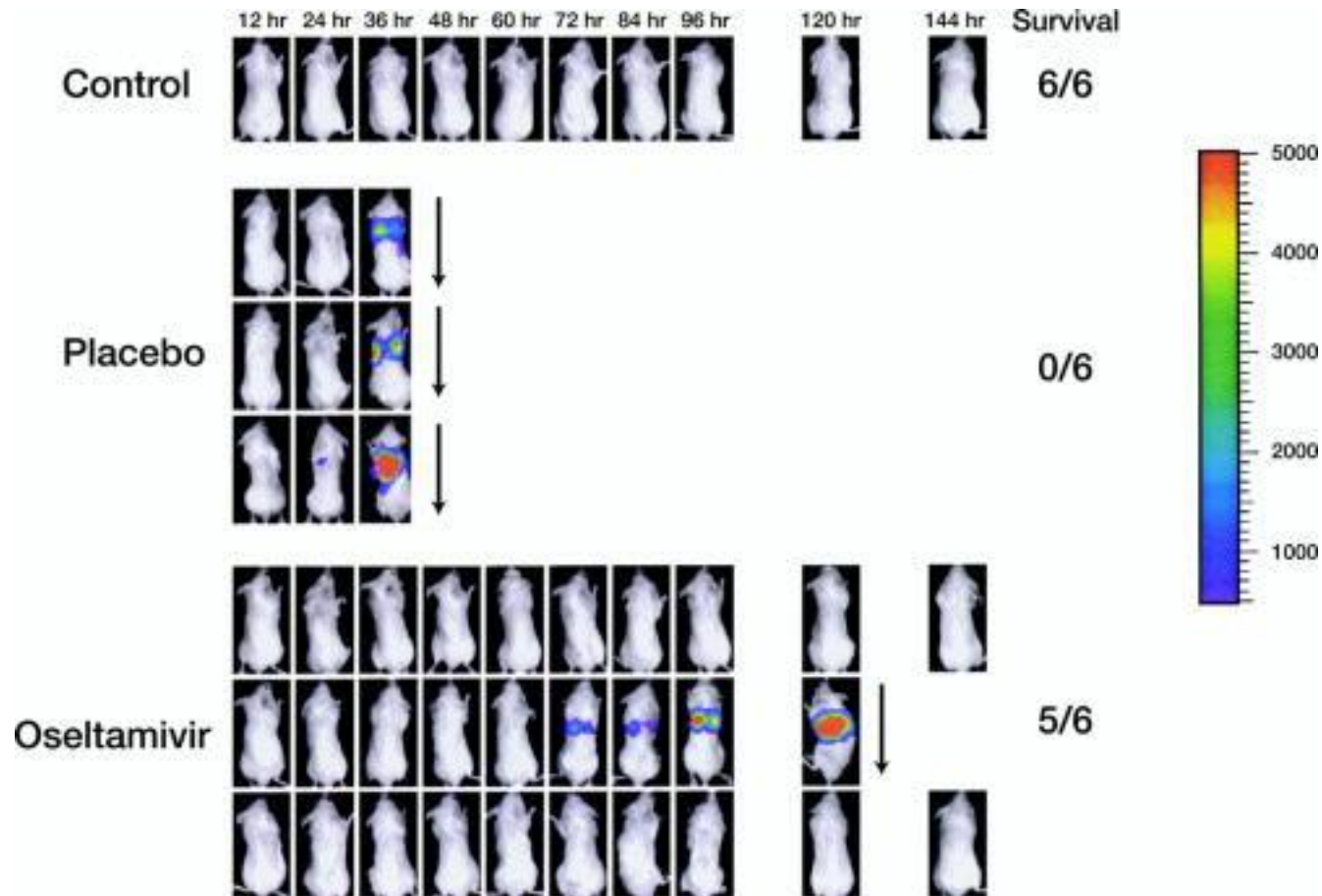
EFFECTS OF HELMINTHS ON HIV

Total number of HIV positive participants	Number infected and helminth species	Effect of co-infection with helminths on CD4 ⁺ T cell count	Effect of co-infection with helminths on viral load	Country (ref)	Comments
111	54 All species (predominant species: <i>Ascaris</i> and hookworm)	Higher ($P = 0.02$)	NS	Zambia (51)	
185	143 Schistosomiasis	NS	Not done	Zimbabwe (38)	Compared subjects with and without schistosomiasis
539	290 All species	NS	Higher ($P = 0.03$) ^a	Uganda (52)	^a (NS after adjustment for potential confounding factors)
	Hookworm	Higher ($P = 0.007$)	NS		
	<i>S. mansoni</i>	NS	NS		
	<i>Strongyloides</i>	NS	Higher ($P = 0.02$) ^a		
	<i>Mansonella</i>	Higher ($P = 0.04$) ^a	NS		
108	39 All species (predominant species: <i>S. mansoni</i> and hookworm)	Higher ($P = 0.005$)	NS	Uganda (53)	
56	31 All species (predominant species: <i>Trichuris</i> and <i>Ascaris</i>)	NS	NS	Ethiopia (54)	
365	161 All species ^b (predominant species: <i>Ascaris</i> , <i>Strongyloides</i> , <i>Trichuris</i> and hookworm)	NS	Lower ($P = 0.02$)	Brazil (42)	^b All species' includes protozoa; information for all helminths without protozoa not given

HIV AND CO-INFECTION



INFLUENZA AND BACTERIA



These effects have to be exquisitely timed: usually 3-7 days after influenza infection for most models

PROPOSED MECHANISMS

- Epithelial damage by the virus (killing, NA activity) creating more hospitable niches for the bacteria
- Innate immune alterations—suppressive environment (corticosteroids, Treg activity, innate recognition refraction)
- Loss of “tolerance”



SCHISTOSOMIASIS MODELING

○ Between-host:

- Chan et al (1995) Epid Inf, Chan et al (1996) Acta Tropica, Chan et al (1996) Am J Trop Med Hyg, French et al (2010) PLoS Neg Trop Dis, Gurarie et al (2005) Parasitology, Liang et al (2002) Acta Tropica, Liang et al (2005) Trop Med Int Health, Liang et al (2005) PNAS, Riley (2008) PloS Med, **Anderson & May (1991) (book), Liang et al (2008) PLoS Med (review), Woolhouse (1991) Acta Tropica (review)**

○ Within-host:

- Chiyaka et al (2010) Comp Math Meth Med, ???

